

Modern Control Engineering

Third Edition

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Prentice Hall, Upper Saddle River, New Jersey 07458

Library of Congress Cataloging-in-Publication Data

Ogata, Katsuhiko.
Modern control engineering / Katsuhiko Ogata. — 3rd ed.
p. cm.
Includes bibliographical references and index.
ISBN: 0-13-227307-1
1. Automatic control. 2. Control theory. I. Title.
TJ213.028 1997
629.8—dc20 96-2345
CIP

Publisher: Tom Robbins
Associate editor: Alice Dworkin
Production editor: Ann Marie Longobardo
Cover designer: Bruce Kenselaar
Manufacturing Buyer: Donna Sullivan



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Simon & Schuster/A Viacom Company
Upper Saddle River, NJ 07458

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Printed in the United States of America

10 9 8

ISBN: 0-13-227307-1

Prentice-Hall International (UK) Limited, London
Prentice-Hall of Australia Pty. Limited, Sydney
Prentice-Hall Canada Inc., Toronto
Prentice-Hall Hispanoamericana, S.A., Mexico
Prentice-Hall of India Private Limited, New Delhi
Prentice-Hall of Japan, Inc., Tokyo
Simon & Schuster Asia Ltd., Singapore
Editora Prentice-Hall do Brasil, Ltda., Rio de Janeiro

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Contents

Preface	ix
Chapter 1 Introduction to Control Systems	1
1-1 Introduction	1
1-2 Examples of Control Systems	3
1-3 Closed-Loop Control Versus Open-Loop Control	6
1-4 Design of Control Systems	8
1-5 Outline of the Book	9
Example Problems and Solutions	10
Problems	11
Chapter 2 The Laplace Transform	13
2-1 Introduction	13
2-2 Review of Complex Variables and Complex Functions	14
2-3 Laplace Transformation	17
2-4 Laplace Transform Theorems	27
2-5 Inverse Laplace Transformation	35
2-6 Partial-Fraction Expansion with MATLAB	41
2-7 Solving Linear, Time-Invariant, Differential Equations	44

Example Problems and Solutions	46
Problems	55

Chapter 3 Mathematical Modeling of Dynamic Systems **57**

3-1	Introduction	57
3-2	Transfer Function and Impulse-Response Function	60
3-3	Block Diagrams	63
3-4	Modeling in State Space	70
3-5	State-Space Representation of Dynamic Systems	76
3-6	Mechanical Systems	81
3-7	Electrical Systems	87
3-8	Liquid-Level Systems	92
3-9	Thermal Systems	96
3-10	Linearization of Nonlinear Mathematical Models	100
	Example Problems and Solutions	105
	Problems	129

Chapter 4 Transient-Response Analysis **134**

4-1	Introduction	134
4-2	First-Order Systems	136
4-3	Second-Order Systems	141
4-4	Transient-Response Analysis with MATLAB	160
4-5	An Example Problem Solved with MATLAB	178
	Example Problems and Solutions	187
	Problems	207

Chapter 5 Basic Control Actions and Response of Control Systems **211**

5-1	Introduction	211
5-2	Basic Control Actions	212
5-3	Effects of Integral and Derivative Control Actions on System Performance	219
5-4	Higher-Order Systems	228
5-5	Routh's Stability Criterion	232
5-6	Pneumatic Controllers	238
5-7	Hydraulic Controllers	255
5-8	Electronic Controllers	262
5-9	Phase Lead and Phase Lag in Sinusoidal Response	269
5-10	Steady-State Errors in Unity-Feedback Control Systems	274

Example Problems and Solutions	282
Problems	309

Chapter 6 Root-Locus Analysis **317**

6-1	Introduction	317
6-2	Root-Locus Plots	319
6-3	Summary of General Rules for Constructing Root Loci	330
6-4	Root-Locus Plots with MATLAB	338
6-5	Special Cases	348
6-6	Root-Locus Analysis of Control Systems	357
6-7	Root Loci for Systems with Transport Lag	360
6-8	Root-Contour Plots	364
	Example Problems and Solutions	368
	Problems	400

Chapter 7 Control Systems Design by the Root-Locus Method **404**

7-1	Introduction	404
7-2	Preliminary Design Considerations	407
7-3	Lead Compensation	409
7-4	Lag Compensation	418
7-5	Lag-Lead Compensation	427
	Example Problems and Solutions	439
	Problems	467

Chapter 8 Frequency-Response Analysis **471**

8-1	Introduction	471
8-2	Bode Diagrams	473
8-3	Plotting Bode Diagrams with MATLAB	492
8-4	Polar Plots	504
8-5	Drawing Nyquist Plots with MATLAB	512
8-6	Log-Magnitude versus Phase Plots	519
8-7	Nyquist Stability Criterion	521
8-8	Stability Analysis	532
8-9	Relative Stability	542
8-10	Closed-Loop Frequency Response	556
8-11	Experimental Determination of Transfer Functions	567
	Example Problems and Solutions	573
	Problems	605

Chapter 9 Control Systems Design By Frequency Response	609
9-1 Introduction	609
9-2 Lead Compensation	612
9-3 Lag Compensation	621
9-4 Lag–Lead Compensation	630
9-5 Concluding Comments	636
Example Problems and Solutions	639
Problems	667
Chapter 10 PID Controls and Introduction to Robust Control	669
10-1 Introduction	669
10-2 Tuning Rules for PID Controllers	670
10-3 Modifications of PID Control Schemes	679
10-4 Two-Degrees-of-Freedom Control	683
10-5 Design Considerations for Robust Control	685
Example Problems and Solutions	690
Problems	703
Chapter 11 Analysis of Control Systems in State Space	710
11-1 Introduction	710
11-2 State-Space Representations of Transfer-Function Systems	711
11-3 Transformation of System Models with MATLAB	718
11-4 Solving The Time-Invariant State Equation	722
11-5 Some Useful Results in Vector-Matrix Analysis	729
11-6 Controllability	737
11-7 Observability	743
Example Problems and Solutions	749
Problems	783
Chapter 12 Design of Control Systems in State Space	786
12-1 Introduction	786
12-2 Pole Placement	787
12-3 Solving Pole-Placement Problems with MATLAB	798
12-4 Design of Regulator-Type Systems by Pole Placement	803
12-5 State Observers	813
12-6 Design of State Observers with MATLAB	837
12-7 Design of Servo Systems	843

12-8 Example of Control System Design with MATLAB	852
Example Problems and Solutions	864
Problems	893
Chapter 13 Liapunov Stability Analysis and Quadratic Optimal Control	896
13-1 Introduction	896
13-2 Liapunov Stability Analysis	897
13-3 Liapunov Stability Analysis of Linear, Time-Invariant Systems	907
13-4 Model-Reference Control Systems	912
13-5 Quadratic Optimal Control	915
13-6 Solving Quadratic Optimal Control Problems with MATLAB	925
Example Problems and Solutions	935
Problems	958
Appendix Background Materials Necessary for the Effective Use of MATLAB	960
A-1 Introduction	960
A-2 Plotting Response Curves	965
A-3 Computing Matrix Functions	967
A-4 Mathematical Models of Linear Systems	977
References	983
Index	987

Preface

This book is written at the level of the senior engineering student and is intended to be used as a text for the first course in control systems. It presents a comprehensive treatment of the analysis and design of continuous-time control systems. It is assumed that the reader has had courses on introductory differential equations, introductory vector-matrix analysis, introductory circuit analysis, and mechanics.

In this third edition, MATLAB® is integrated into the text. All computational problems are solved with MATLAB. Also, design aspects have been strengthened, and new topics, examples and problems are added.

The text is organized into 13 chapters and an appendix. The outline of the book is as follows: Chapter 1 presents introductory materials on control systems. Chapter 2 gives the Laplace transforms of commonly encountered time functions and basic Laplace transform theorems. (If the students have an adequate background on the Laplace transform, this chapter may be skipped.) Chapter 3 treats mathematical modeling of dynamic systems and develops transfer-function models and state-space models. Chapter 4 gives transient-response analysis of first- and second-order systems. This chapter includes a computational analysis of transient response by use of MATLAB. Chapter 5 presents basic control actions in industrial automatic controllers and discusses pneumatic, hydraulic, and electronic controllers. This chapter also discusses the response of higher-order systems and Routh's stability criterion.

Chapter 6 treats the root-locus analysis. The MATLAB approach to plotting root loci is presented in this chapter. Chapter 7 presents the design of lead, lag and lag-lead compensators by the root-locus method. Chapter 8 deals with the frequency-response analysis of control systems. Bode diagrams, polar plots, the Nyquist stability criterion,

and closed-loop frequency response are discussed, including the MATLAB approach to obtain frequency-response plots. Chapter 9 covers the design and compensation techniques using frequency-response methods. Specifically, the Bode diagram approach to the design of lead, lag, and lag-lead compensators is discussed in detail in this chapter. Chapter 10 deals with the basic and modified PID controls. This chapter gives discussions of two-degrees-of-freedom controls and design considerations for robust control.

Chapter 11 presents a basic analysis of control systems in state space. Concepts of controllability and observability are given here. The transformation of system models (from transfer-function model to state-space model, and vice versa) by the use of MATLAB is included in this chapter. Chapter 12 treats the design of control systems in state space. This chapter begins with pole-placement design problems, followed by the design of state observers. A design of a type 1 servo system based on the pole-placement approach is presented, including a computational solution with MATLAB. Chapter 13 begins with Liapunov stability analysis, followed by design of a model-reference control system, where the conditions for Liapunov stability are formulated first and then the system is designed within these limitations. Then quadratic optimal control problems are treated. Here the Liapunov stability equation is used to lead into quadratic optimal control theory. A MATLAB solution to the quadratic optimal control problem is also presented.

No prior knowledge of MATLAB is assumed in this book. If the reader is not yet familiar with MATLAB, it is suggested that he or she read the appendix first and then study MATLAB as presented in the text.

Throughout the book the basic concepts involved are emphasized and highly mathematical arguments are carefully avoided in the presentation of the materials. Mathematical proofs are provided when they contribute to the understanding of the subjects presented. All the material has been organized toward a gradual development of control theory.

Examples are presented at strategic points throughout the book so that the reader will have a better understanding of the subject matter discussed. In addition, a number of solved problems (A-problems) are provided at the end of each chapter. These problems constitute an integral part of the text. It is suggested that the reader study all of these problems carefully to obtain a deeper understanding of the topics discussed. In addition, many unsolved problems (B-problems) are provided for use as homework or quiz problems. An instructor using this text can obtain a complete solutions manual (for B-problems) from the publisher.

Most of the materials presented in this book have been class tested in senior and first-year graduate-level courses on control systems at the University of Minnesota.

If this book is used as a text for a four-hour quarter course (with 40 lecture hours) or a three-hour semester course (with 42 lecture hours), most of the materials in the first 10 chapters may be covered. (The first 10 chapters cover all basic materials of control systems normally required in a first course on control systems.) If this book is used as a text for a four-hour semester course (with 52 lecture hours), a good part of the book may be covered with flexibility in omitting certain subjects. For a two-quarter course sequence (with 60 or more lecture hours), the entire book may be covered. This book can also serve as a self-study book for practicing engineers who wish to study basic materials of control theory.

I would like to express my sincere appreciation to Professor Suhada Jayasuriya of Texas A & M University, who reviewed the final manuscript and made many constructive comments. Appreciation is also due to Linda Ratts Engelman for her enthusiasm in publishing the third edition, to the anonymous reviewers who made valuable suggestions at the early stage of the revision process, and to my former students who solved many of the A-problems and B-problems included in this book.

Katsuhiko Ogata

1

Introduction to Control Systems

1-1 INTRODUCTION

Automatic control has played a vital role in the advance of engineering and science. In addition to its extreme importance in space-vehicle systems, missile-guidance systems, robotic systems, and the like, automatic control has become an important and integral part of modern manufacturing and industrial processes. For example, automatic control is essential in the numerical control of machine tools in the manufacturing industries, in the design of autopilot systems in the aerospace industries, and in the design of cars and trucks in the automobile industries. It is also essential in such industrial operations as controlling pressure, temperature, humidity, viscosity, and flow in the process industries.

Since advances in the theory and practice of automatic control provide the means for attaining optimal performance of dynamic systems, improving productivity, relieving the drudgery of many routine repetitive manual operations, and more, most engineers and scientists must now have a good understanding of this field.

Historical review. The first significant work in automatic control was James Watt's centrifugal governor for the speed control of a steam engine in the eighteenth century. Other significant works in the early stages of development of control theory were due to Minorsky, Hazen, and Nyquist, among many others. In 1922, Minorsky worked on automatic controllers for steering ships and showed how stability could be determined from the differential equations describing the system. In 1932, Nyquist developed a relatively simple procedure for determining the stability of closed-loop systems on the basis of open-loop response to steady-state sinusoidal inputs. In 1934, Hazen, who

introduced the term servomechanisms for position control systems, discussed the design of relay servomechanisms capable of closely following a changing input.

During the decade of the 1940s, frequency-response methods made it possible for engineers to design linear closed-loop control systems that satisfied performance requirements. From the end of the 1940s to early 1950s, the root-locus method due to Evans was fully developed.

The frequency-response and root-locus methods, which are the core of classical control theory, lead to systems that are stable and satisfy a set of more or less arbitrary performance requirements. Such systems are, in general, acceptable but not optimal in any meaningful sense. Since the late 1950s, the emphasis in control design problems has been shifted from the design of one of many systems that work to the design of one optimal system in some meaningful sense.

As modern plants with many inputs and outputs become more and more complex, the description of a modern control system requires a large number of equations. Classical control theory, which deals only with single-input–single-output systems, becomes powerless for multiple-input–multiple-output systems. Since about 1960, because the availability of digital computers made possible time-domain analysis of complex systems, modern control theory, based on time-domain analysis and synthesis using state variables, has been developed to cope with the increased complexity of modern plants and the stringent requirements on accuracy, weight, and cost in military, space, and industrial applications.

During the years from 1960 to 1980, optimal control of both deterministic and stochastic systems, as well as adaptive and learning control of complex systems, were fully investigated. From 1980 to the present, developments in modern control theory centered around robust control, H_∞ control, and associated topics.

Now that digital computers have become cheaper and more compact, they are used as integral parts of control systems. Recent applications of modern control theory include such nonengineering systems as biological, biomedical, economic, and socio-economic systems.

Definitions. Before we can discuss control systems, some basic terminologies must be defined.

Controlled Variable and Manipulated Variable. The *controlled* variable is the quantity or condition that is measured and controlled. The *manipulated* variable is the quantity or condition that is varied by the controller so as to affect the value of the controlled variable. Normally, the controlled variable is the output of the system. *Control* means measuring the value of the controlled variable of the system and applying the manipulated variable to the system to correct or limit deviation of the measured value from a desired value.

In studying control engineering, we need to define additional terms that are necessary to describe control systems.

Plants. A plant may be a piece of equipment, perhaps just a set of machine parts functioning together, the purpose of which is to perform a particular operation. In this book, we shall call any physical object to be controlled (such as a mechanical device, a heating furnace, a chemical reactor, or a spacecraft) a plant.

Processes. The *Merriam-Webster Dictionary* defines a process to be a natural, progressively continuing operation or development marked by a series of gradual changes that succeed one another in a relatively fixed way and lead toward a particular result or end; or an artificial or voluntary, progressively continuing operation that consists of a series of controlled actions or movements systematically directed toward a particular result or end. In this book we shall call any operation to be controlled a *process*. Examples are chemical, economic, and biological processes.

Systems. A system is a combination of components that act together and perform a certain objective. A system is not limited to physical ones. The concept of the system can be applied to abstract, dynamic phenomena such as those encountered in economics. The word system should, therefore, be interpreted to imply physical, biological, economic, and the like, systems.

Disturbances. A disturbance is a signal that tends to adversely affect the value of the output of a system. If a disturbance is generated within the system, it is called *internal*, while an *external* disturbance is generated outside the system and is an input.

Feedback Control. Feedback control refers to an operation that, in the presence of disturbances, tends to reduce the difference between the output of a system and some reference input and that does so on the basis of this difference. Here only unpredictable disturbances are so specified, since predictable or known disturbances can always be compensated for within the system.

1-2 EXAMPLES OF CONTROL SYSTEMS

In this section we shall present several examples of control systems.

Speed control system. The basic principle of a Watt's speed governor for an engine is illustrated in the schematic diagram of Figure 1-1. The amount of fuel admitted

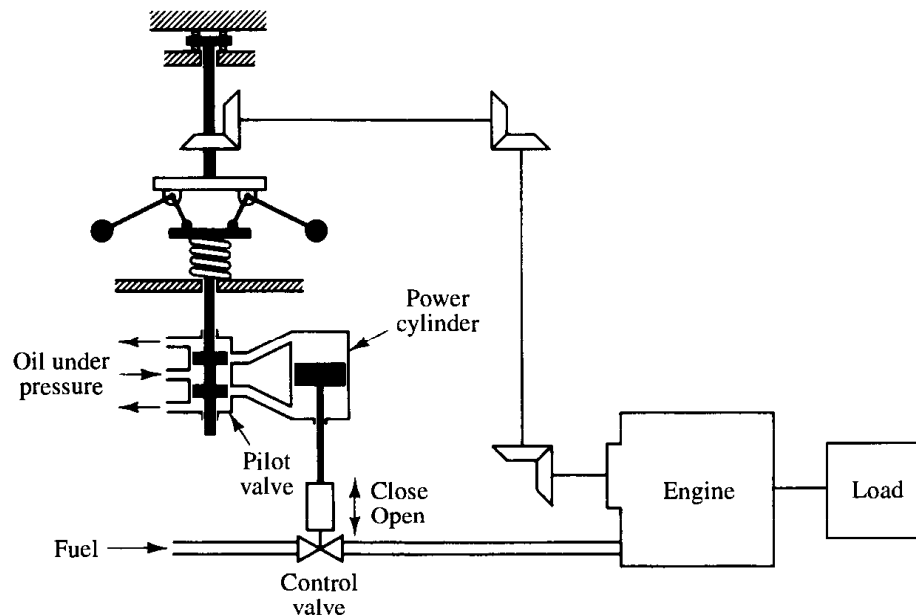


Figure 1-1
Speed control system.

to the engine is adjusted according to the difference between the desired and the actual engine speeds.

The sequence of actions may be stated as follows: The speed governor is adjusted such that, at the desired speed, no pressured oil will flow into either side of the power cylinder. If the actual speed drops below the desired value due to disturbance, then the decrease in the centrifugal force of the speed governor causes the control valve to move downward, supplying more fuel, and the speed of the engine increases until the desired value is reached. On the other hand, if the speed of the engine increases above the desired value, then the increase in the centrifugal force of the governor causes the control valve to move upward. This decreases the supply of fuel, and the speed of the engine decreases until the desired value is reached.

In this speed control system, the plant (controlled system) is the engine and the controlled variable is the speed of the engine. The difference between the desired speed and the actual speed is the error signal. The control signal (the amount of fuel) to be applied to the plant (engine) is the actuating signal. The external input to disturb the controlled variable is the disturbance. An unexpected change in the load is a disturbance.

Robot control system. Industrial robots are frequently used in industry to improve productivity. The robot can handle monotonous jobs as well as complex jobs without errors in operation. The robot can work in an environment intolerable to human operators. For example, it can work in extreme temperatures (both high and low) or in a high- or low-pressure environment or under water or in space. There are special robots for fire fighting, underwater exploration, and space exploration, among many others.

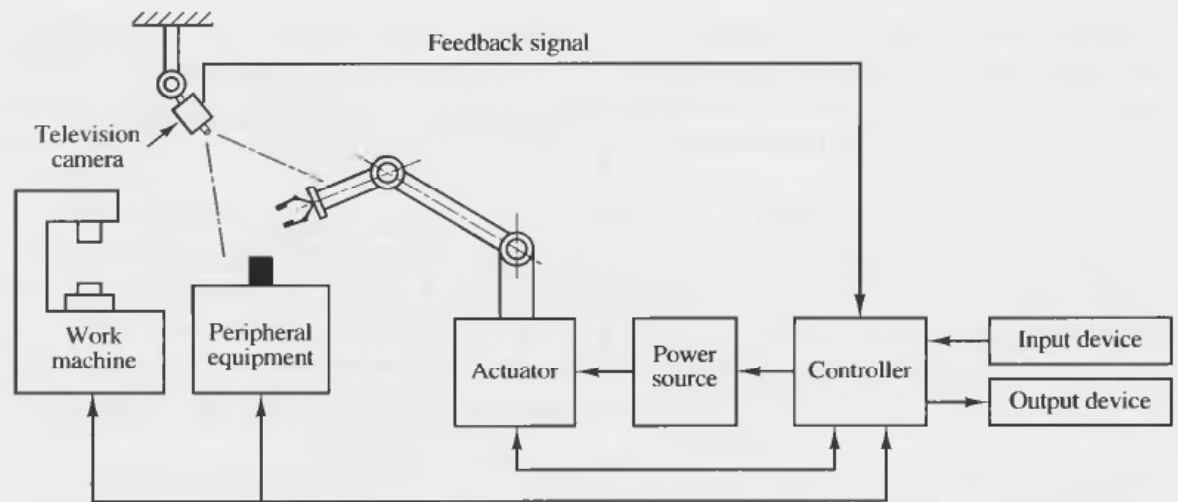
The industrial robot must handle mechanical parts that have particular shapes and weights. Hence, it must have at least an arm, a wrist, and a hand. It must have sufficient power to perform the task and the capability for at least limited mobility. In fact, some robots of today are able to move freely by themselves in a limited space in a factory.

The industrial robot must have some sensory devices. In low-level robots, microswitches are installed in the arms as sensory devices. The robot first touches an object and then, through the microswitches, confirms the existence of the object in space and proceeds in the next step to grasp it.

In a high-level robot, an optical means (such as a television system) is used to scan the background of the object. It recognizes the pattern and determines the presence and orientation of the object. A computer is necessary to process signals in the pattern-recognition process (see Figure 1–2). In some applications, the computerized robot recognizes the presence and orientation of each mechanical part by a pattern-recognition process that consists of reading the code numbers attached to it. Then the robot picks up the part and moves it to an appropriate place for assembling, and there it assembles several parts into a component. A well-programmed digital computer acts as a controller.

Temperature control system. Figure 1–3 shows a schematic diagram of temperature control of an electric furnace. The temperature in the electric furnace is measured by a thermometer, which is an analog device. The analog temperature is converted to a digital temperature by an A/D converter. The digital temperature is fed to a controller through an interface. This digital temperature is compared with the programmed input temperature, and if there is any discrepancy (error), the controller sends out a signal to

Figure 1-2
Robot using a
pattern-recognition
process.



the heater, through an interface, amplifier, and relay, to bring the furnace temperature to a desired value.

Temperature control of the passenger compartment of a car. Figure 1-4 shows a functional diagram of temperature control of the passenger compartment of a car. The desired temperature, converted to a voltage, is the input to the controller. The actual temperature of the passenger compartment is converted to a voltage through a sensor and is fed back to the controller for comparison with the input. The ambient temperature and radiation heat transfer from the sun, which are not constant while the car is driven, act as disturbances. This system employs both feedback control and feedforward control. (Feedforward control gives corrective action before the disturbances affect the output.)

The temperature of the passenger car compartment differs considerably depending on the place where it is measured. Instead of using multiple sensors for temperature measurement and averaging the measured values, it is economical to install a small suction blower at the place where passengers normally sense the temperature. The temperature of the air from the suction blower is an indication of the passenger compartment temperature and is considered the output of the system.

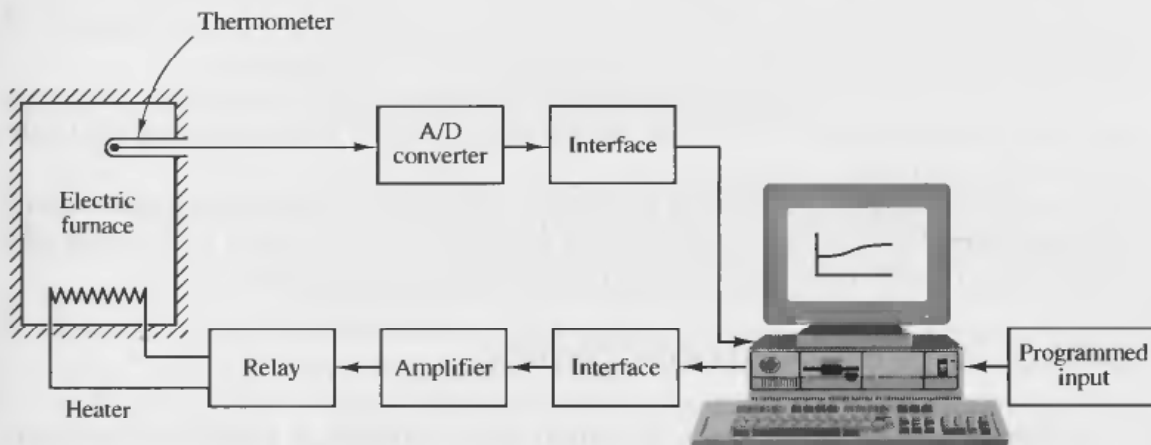


Figure 1-3
Temperature control system.

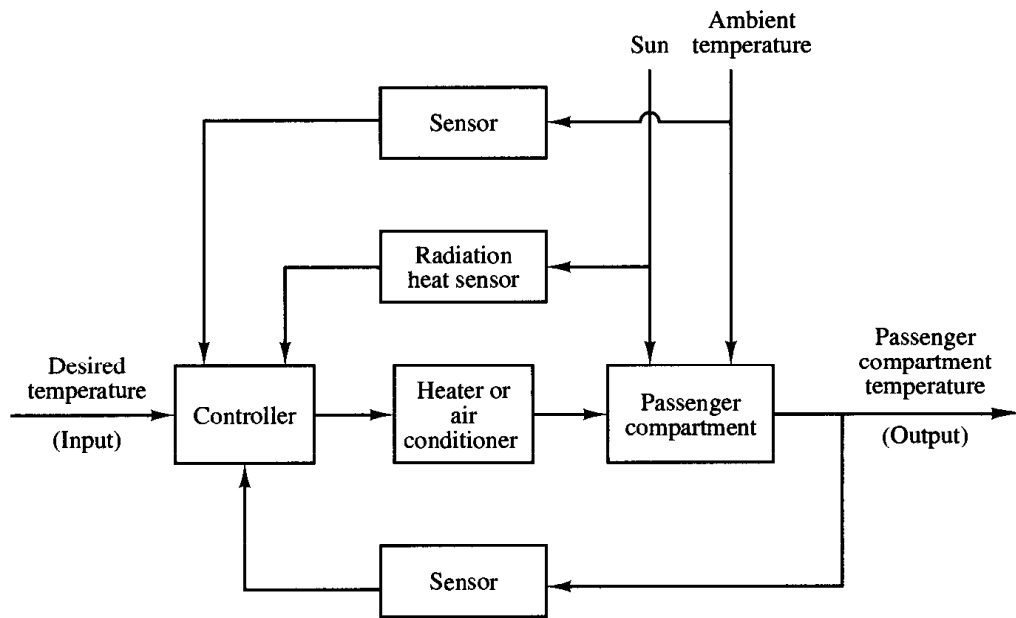


Figure 1-4
Temperature control
of passenger com-
partment of a car.

The controller receives the input signal, output signal, and signals from sensors from disturbance sources. The controller sends out an optimal control signal to the air conditioner or heater to control the amount of cooling air or warm air so that the passenger compartment temperature is about the desired temperature.

Business systems. A business system may consist of many groups. Each task assigned to a group will represent a dynamic element of the system. Feedback methods of reporting the accomplishments of each group must be established in such a system for proper operation. The cross-coupling between functional groups must be made a minimum in order to reduce undesirable delay times in the system. The smaller this cross-coupling, the smoother the flow of work signals and materials will be.

A business system is a closed-loop system. A good design will reduce the managerial control required. Note that disturbances in this system are the lack of personnel or materials, interruption of communication, human errors, and the like.

The establishment of a well-founded estimating system based on statistics is mandatory to proper management. (Note that it is a well-known fact that the performance of such a system can be improved by the use of lead time, or *anticipation*.)

To apply control theory to improve the performance of such a system, we must represent the dynamic characteristic of the component groups of the system by a relatively simple set of equations.

Although it is certainly a difficult problem to derive mathematical representations of the component groups, the application of optimization techniques to business systems significantly improves the performance of the business system.

1-3 CLOSED-LOOP CONTROL VERSUS OPEN-LOOP CONTROL

Feedback control systems. A system that maintains a prescribed relationship between the output and the reference input by comparing them and using the difference as a means of control is called a *feedback control system*. An example would

be a room-temperature control system. By measuring the actual room temperature and comparing it with the reference temperature (desired temperature), the thermostat turns the heating or cooling equipment on or off in such a way as to ensure that the room temperature remains at a comfortable level regardless of outside conditions.

Feedback control systems are not limited to engineering but can be found in various nonengineering fields as well. The human body, for instance, is a highly advanced feedback control system. Both body temperature and blood pressure are kept constant by means of physiological feedback. In fact, feedback performs a vital function: It makes the human body relatively insensitive to external disturbances, thus enabling it to function properly in a changing environment.

Closed-loop control systems. Feedback control systems are often referred to as *closed-loop control systems*. In practice, the terms feedback control and closed-loop control are used interchangeably. In a closed-loop control system the actuating error signal, which is the difference between the input signal and the feedback signal (which may be the output signal itself or a function of the output signal and its derivatives and/or integrals), is fed to the controller so as to reduce the error and bring the output of the system to a desired value. The term closed-loop control always implies the use of feedback control action in order to reduce system error.

Open-loop control systems. Those systems in which the output has no effect on the control action are called *open-loop control systems*. In other words, in an open-loop control system the output is neither measured nor fed back for comparison with the input. One practical example is a washing machine. Soaking, washing, and rinsing in the washer operate on a time basis. The machine does not measure the output signal, that is, the cleanliness of the clothes.

In any open-loop control system the output is not compared with the reference input. Thus, to each reference input there corresponds a fixed operating condition; as a result, the accuracy of the system depends on calibration. In the presence of disturbances, an open-loop control system will not perform the desired task. Open-loop control can be used, in practice, only if the relationship between the input and output is known and if there are neither internal nor external disturbances. Clearly, such systems are not feedback control systems. Note that any control system that operates on a time basis is open loop. For instance, traffic control by means of signals operated on a time basis is another example of open-loop control.

Closed-loop versus open-loop control systems. An advantage of the closed-loop control system is the fact that the use of feedback makes the system response relatively insensitive to external disturbances and internal variations in system parameters. It is thus possible to use relatively inaccurate and inexpensive components to obtain the accurate control of a given plant, whereas doing so is impossible in the open-loop case.

From the point of view of stability, the open-loop control system is easier to build because system stability is not a major problem. On the other hand, stability is a major problem in the closed-loop control system, which may tend to overcorrect errors that can cause oscillations of constant or changing amplitude.

It should be emphasized that for systems in which the inputs are known ahead of time and in which there are no disturbances it is advisable to use open-loop control. Closed-loop control systems have advantages only when unpredictable disturbances and/or unpredictable variations in system components are present. Note that the output power rating partially determines the cost, weight, and size of a control system. The number of components used in a closed-loop control system is more than that for a corresponding open-loop control system. Thus, the closed-loop control system is generally higher in cost and power. To decrease the required power of a system, open-loop control may be used where applicable. A proper combination of open-loop and closed-loop controls is usually less expensive and will give satisfactory overall system performance.

1-4 DESIGN OF CONTROL SYSTEMS

Actual control systems are generally nonlinear. However, if they can be approximated by linear mathematical models, we may use one or more of the well-developed design methods. In a practical sense, the performance specifications given to the particular system suggest which method to use. If the performance specifications are given in terms of transient-response characteristics and/or frequency-domain performance measures, then we have no choice but to use a conventional approach based on the root-locus and/or frequency-response methods. (These methods are presented in Chapters 6 through 9.) If the performance specifications are given as performance indexes in terms of state variables, then modern control approaches should be used. (These approaches are presented in Chapters 11 through 13.)

While control system design via the root-locus and frequency-response approaches is an engineering endeavor, system design in the context of modern control theory (state-space methods) employs mathematical formulations of the problem and applies mathematical theory to design problems in which the system can have multiple inputs and multiple outputs and can be time varying. By applying modern control theory, the designer is able to start from a performance index, together with constraints imposed on the system, and to proceed to design a stable system by a completely analytical procedure. The advantage of design based on such modern control theory is that it enables the designer to produce a control system that is optimal with respect to the performance index considered.

The systems that may be designed by a conventional approach are usually limited to single-input-single-output, linear time-invariant systems. The designer seeks to satisfy all performance specifications by means of educated trial-and-error repetition. After a system is designed, the designer checks to see if the designed system satisfies all the performance specifications. If it does not, then he repeats the design process by adjusting parameter settings or by changing the system configuration until the given specifications are met. Although the design is based on a trial-and-error procedure, the ingenuity and know-how of the designer will play an important role in a successful design. An experienced designer may be able to design an acceptable system without using many trials.

It is generally desirable that the designed system should exhibit as small errors as possible in responding to the input signal. In this regard, the damping of the system should be reasonable. The system dynamics should be relatively insensitive to small changes in system parameters. The undesirable disturbances should be well attenuated. [In general, the high-frequency portion should attenuate fast so that high-frequency noises (such as sensor noises) can be attenuated. If the noise or disturbance frequencies are known, notch filters may be used to attenuate these specific frequencies.] If the design of the system is boiled down to a few candidates, an optimal choice among them may be made from such considerations as projected overall performance, cost, space, and weight.

1-5 OUTLINE OF THE BOOK

In what follows we shall briefly present the arrangements and contents of the book.

Chapter 1 has given introductory materials on control systems. Chapter 2 presents basic Laplace transform theory necessary for understanding the control theory presented in this book. Chapter 3 deals with mathematical modeling of dynamic systems in terms of transfer functions and state-space equations. This chapter includes discussions of linearization of nonlinear systems. Chapter 4 treats transient-response analyses of first- and second-order systems. This chapter also gives details of transient-response analysis with MATLAB. Chapter 5 first presents basic control actions and then discusses pneumatic, hydraulic, and electronic controllers. This chapter also discusses Routh's stability criterion.

Chapter 6 gives a root-locus analysis of control systems. General rules for constructing root loci are presented. Detailed discussions for plotting root loci with MATLAB are included. Chapter 7 deals with the design of control systems via the root-locus method. Specifically, root-locus approaches to the design of lead compensators, lag compensators, and lag-lead compensators are discussed in detail. Chapter 8 gives the frequency-response analysis of control systems. Bode diagrams, polar plots, Nyquist stability criterion, and closed-loop frequency response are discussed. Chapter 9 treats control systems design via the frequency-response approach. Here Bode diagrams are used to design lead compensators, lag compensators, and lag-lead compensators. Chapter 10 discusses the basic and modified PID controls. Topics included are tuning rules for PID controllers, modifications of PID control schemes, two-degrees-of-freedom control, and design considerations for robust control.

Chapter 11 presents basic materials for the state-space analysis of control systems. The solution of the time-invariant state equation is derived and concepts of controllability and observability are discussed. Chapter 12 treats the design of control systems in state space. This chapter begins with the pole-placement problems, followed by the design of state observers, and concludes with the design of type 1 servo systems. MATLAB is utilized in solving pole-placement problems, design of state observers, and design of servo systems. Chapter 13, the final chapter, presents the Liapunov stability analysis and the quadratic optimal control. This chapter begins with the Liapunov stability analysis. Then the Liapunov stability approach is used for designing

model-reference control systems. Finally, quadratic optimal control problems are discussed in detail. Here the Liapunov stability approach is utilized to derive the Riccati equation for quadratic optimal control. MATLAB solutions to quadratic optimal control problems are included.

The appendix summarizes background materials necessary for the effective use of MATLAB. This appendix is specifically provided for those readers who are as yet unfamiliar with MATLAB.

EXAMPLE PROBLEMS AND SOLUTIONS

A-1-1. List the major advantages and disadvantages of open-loop control systems.

Solution. The advantages of open-loop control systems are as follows:

1. Simple construction and ease of maintenance.
2. Less expensive than a corresponding closed-loop system.
3. There is no stability problem.
4. Convenient when output is hard to measure or economically not feasible. (For example, in the washer system, it would be quite expensive to provide a device to measure the quality of the output, cleanliness of the clothes, of the washer.)

The disadvantages of open-loop control systems are as follows:

1. Disturbances and changes in calibration cause errors, and the output may be different from what is desired.
2. To maintain the required quality in the output, recalibration is necessary from time to time.

A-1-2. Figure 1-5(a) is a schematic diagram of a liquid-level control system. Here the automatic controller maintains the liquid level by comparing the actual level with a desired level and correcting any error by adjusting the opening of the pneumatic valve. Figure 1-5(b) is a block diagram of the control system. Draw the corresponding block diagram for a human-operated liquid-level control system.

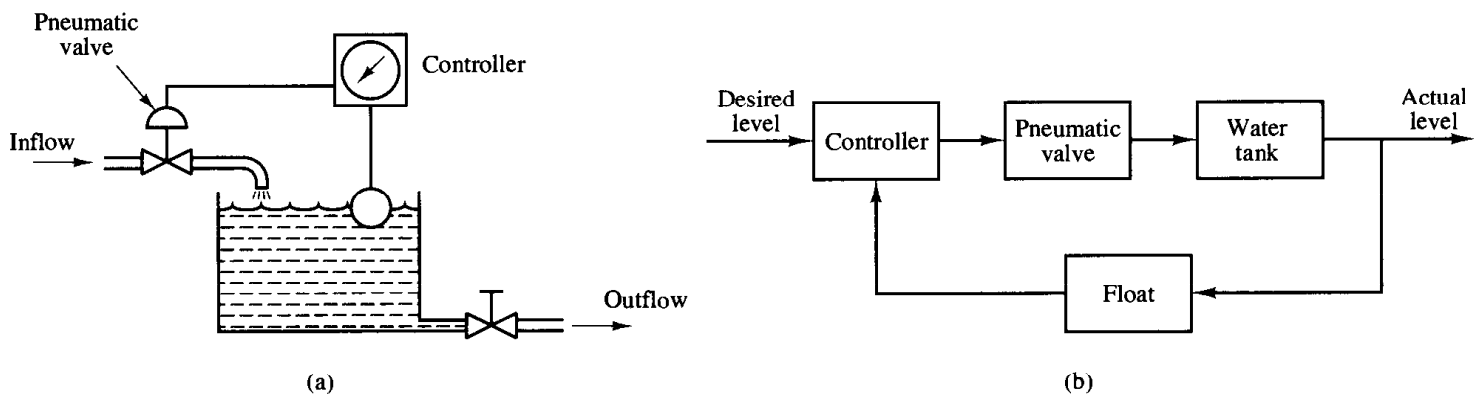
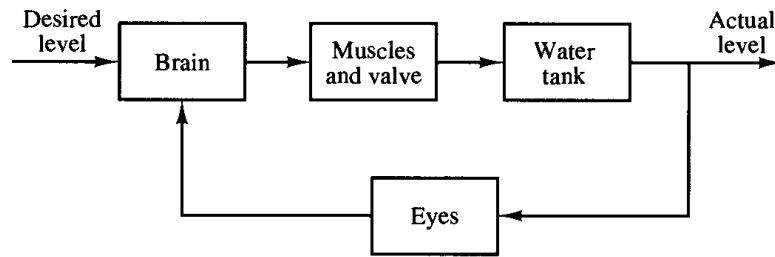


Figure 1-5

(a) Liquid-level control system; (b) block diagram.

Figure 1-6
Block diagram of
human-operated
liquid-level control
system.



Solution. In the human-operated system, the eyes, brain, and muscles correspond to the sensor, controller, and pneumatic valve, respectively. A block diagram is shown in Figure 1-6.

A-1-3. An engineering organizational system is composed of major groups, such as management, research and development, preliminary design, experiments, product design and drafting, fabrication and assembling, and testing. These groups are interconnected to make up the whole operation.

The system may be analyzed by reducing it to the most elementary set of components necessary that can provide the analytical detail required and by representing the dynamic characteristics of each component by a set of simple equations. (The dynamic performance of such a system may be determined from the relation between progressive accomplishment and time.)

Draw a functional block diagram showing an engineering organizational system.

Solution. A functional block diagram can be drawn by using blocks to represent the functional activities and interconnecting signal lines to represent the information or product output of the system operation. A possible block diagram is shown in Figure 1-7.

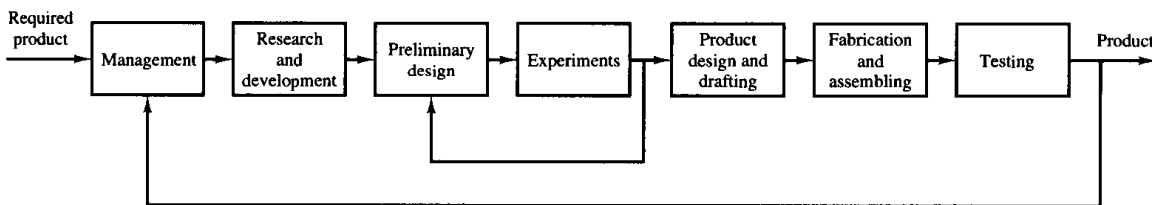


Figure 1-7
Block diagram of an engineering organizational system.

PROBLEMS

B-1-1. Many closed-loop and open-loop control systems may be found in homes. List several examples and describe them.

B-1-2. Give two examples of feedback control systems in which a human acts as a controller.

B-1-3. Figure 1-8 shows a tension control system. Explain the sequence of control actions when the feed speed is suddenly changed for a short time.

B-1-4. Many machines, such as lathes, milling machines, and grinders, are provided with tracers to reproduce the contour of templates. Figure 1-9 shows a schematic diagram of a tracing system in which the tool duplicates the shape of the template on the workpiece. Explain the operation of this system.

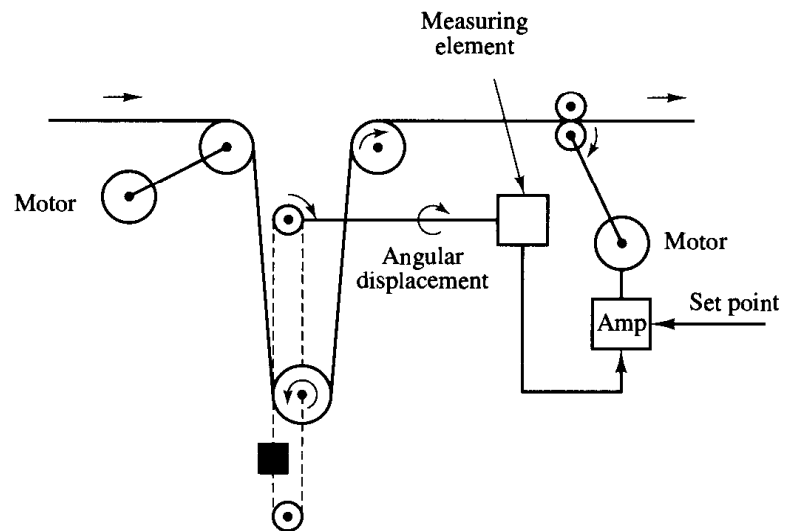


Figure 1–8
Tension control
system.

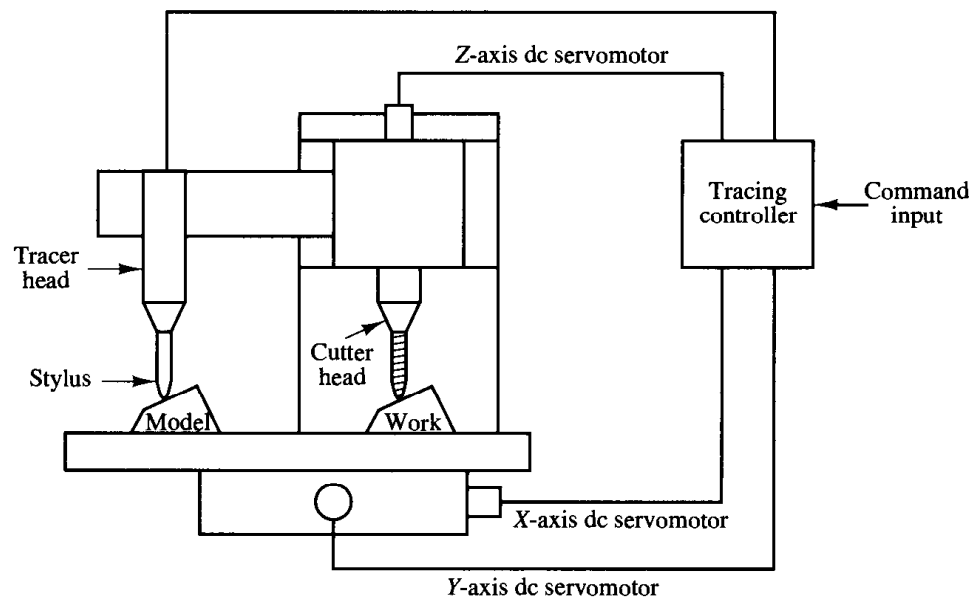


Figure 1–9
Schematic diagram
of a tracing system.

2

The Laplace Transform*

2-1 INTRODUCTION

The Laplace transform method is an operational method that can be used advantageously for solving linear differential equations. By use of Laplace transforms, we can convert many common functions, such as sinusoidal functions, damped sinusoidal functions, and exponential functions, into algebraic functions of a complex variable s . Operations such as differentiation and integration can be replaced by algebraic operations in the complex plane. Thus, a linear differential equation can be transformed into an algebraic equation in a complex variable s . If the algebraic equation in s is solved for the dependent variable, then the solution of the differential equation (the inverse Laplace transform of the dependent variable) may be found by use of a Laplace transform table or by use of the partial-fraction expansion technique, which is presented in Section 2-5.

An advantage of the Laplace transform method is that it allows the use of graphical techniques for predicting the system performance without actually solving system differential equations. Another advantage of the Laplace transform method is that, when we solve the differential equation, both the transient component and steady-state component of the solution can be obtained simultaneously.

Outline of the chapter. Section 2-1 presents introductory remarks. Section 2-2 briefly reviews complex variables and complex functions. Section 2-3 derives Laplace

*This chapter may be skipped if the student is already familiar with Laplace transforms.

transforms of time functions that are frequently used in control engineering. Section 2–4 presents useful theorems of Laplace transforms, and Section 2–5 treats the inverse Laplace transformation. Section 2–6 presents the MATLAB approach to obtain partial-fraction expansion of $B(s)/A(s)$, where $A(s)$ and $B(s)$ are polynomials in s . Finally, Section 2–7 deals with solutions of linear time-invariant differential equations by the Laplace transform approach.

2–2 REVIEW OF COMPLEX VARIABLES AND COMPLEX FUNCTIONS

Before we present the Laplace transformation, we shall review the complex variable and complex function. We shall also review Euler's theorem, which relates the sinusoidal functions to exponential functions.

Complex variable. A complex number has a real part and an imaginary part, both of which are constant. If the real part and/or imaginary part are variables, a complex number is called a *complex variable*. In the Laplace transformation we use the notation s as a complex variable; that is,

$$s = \sigma + j\omega$$

where σ is the real part and ω is the imaginary part.

Complex function. A complex function $F(s)$, a function of s , has a real part and an imaginary part or

$$F(s) = F_x + jF_y$$

where F_x and F_y are real quantities. The magnitude of $F(s)$ is $\sqrt{F_x^2 + F_y^2}$, and the angle θ of $F(s)$ is $\tan^{-1}(F_y/F_x)$. The angle is measured counterclockwise from the positive real axis. The complex conjugate of $F(s)$ is $\bar{F}(s) = F_x - jF_y$.

Complex functions commonly encountered in linear control systems analysis are single-valued functions of s and are uniquely determined for a given value of s .

A complex function $G(s)$ is said to be *analytic* in a region if $G(s)$ and all its derivatives exist in that region. The derivative of an analytic function $G(s)$ is given by

$$\frac{d}{ds} G(s) = \lim_{\Delta s \rightarrow 0} \frac{G(s + \Delta s) - G(s)}{\Delta s} = \lim_{\Delta s \rightarrow 0} \frac{\Delta G}{\Delta s}$$

Since $\Delta s = \Delta\sigma + j\Delta\omega$, Δs can approach zero along an infinite number of different paths. It can be shown, but is stated without a proof here, that if the derivatives taken along two particular paths, that is, $\Delta s = \Delta\sigma$ and $\Delta s = j\Delta\omega$, are equal, then the derivative is unique for any other path $\Delta s = \Delta\sigma + j\Delta\omega$ and so the derivative exists.

For a particular path $\Delta s = \Delta\sigma$ (which means that the path is on the real axis),

$$\frac{d}{ds} G(s) = \lim_{\Delta\sigma \rightarrow 0} \left(\frac{\Delta G_x}{\Delta\sigma} + j \frac{\Delta G_y}{\Delta\sigma} \right) = \frac{\partial G_x}{\partial \sigma} + j \frac{\partial G_y}{\partial \sigma}$$

For another particular path $\Delta s = j\Delta\omega$ (which means that the path is on the imaginary axis),

$$\frac{d}{ds} G(s) = \lim_{j\Delta\omega \rightarrow 0} \left(\frac{\Delta G_x}{j\Delta\omega} + j \frac{\Delta G_y}{j\Delta\omega} \right) = -j \frac{\partial G_x}{\partial \omega} + \frac{\partial G_y}{\partial \omega}$$

If these two values of the derivative are equal,

$$\frac{\partial G_x}{\partial \sigma} + j \frac{\partial G_y}{\partial \sigma} = \frac{\partial G_y}{\partial \omega} - j \frac{\partial G_x}{\partial \omega}$$

or if the following two conditions

$$\frac{\partial G_x}{\partial \sigma} = \frac{\partial G_y}{\partial \omega} \quad \text{and} \quad \frac{\partial G_y}{\partial \sigma} = -\frac{\partial G_x}{\partial \omega}$$

are satisfied, then the derivative $dG(s)/ds$ is uniquely determined. These two conditions are known as the Cauchy–Riemann conditions. If these conditions are satisfied, the function $G(s)$ is analytic.

As an example, consider the following $G(s)$:

$$G(s) = \frac{1}{s + 1}$$

Then

$$G(\sigma + j\omega) = \frac{1}{\sigma + j\omega + 1} = G_x + jG_y$$

where

$$G_x = \frac{\sigma + 1}{(\sigma + 1)^2 + \omega^2} \quad \text{and} \quad G_y = \frac{-\omega}{(\sigma + 1)^2 + \omega^2}$$

It can be seen that, except at $s = -1$ (that is, $\sigma = -1, \omega = 0$), $G(s)$ satisfies the Cauchy–Riemann conditions:

$$\begin{aligned} \frac{\partial G_x}{\partial \sigma} &= \frac{\partial G_y}{\partial \omega} = \frac{\omega^2 - (\sigma + 1)^2}{[(\sigma + 1)^2 + \omega^2]^2} \\ \frac{\partial G_y}{\partial \sigma} &= -\frac{\partial G_x}{\partial \omega} = \frac{2\omega(\sigma + 1)}{[(\sigma + 1)^2 + \omega^2]^2} \end{aligned}$$

Hence $G(s) = 1/(s + 1)$ is analytic in the entire s plane except at $s = -1$. The derivative $dG(s)/ds$, except at $s = -1$, is found to be

$$\begin{aligned} \frac{d}{ds} G(s) &= \frac{\partial G_x}{\partial \sigma} + j \frac{\partial G_y}{\partial \sigma} = \frac{\partial G_y}{\partial \omega} - j \frac{\partial G_x}{\partial \omega} \\ &= -\frac{1}{(\sigma + j\omega + 1)^2} = -\frac{1}{(s + 1)^2} \end{aligned}$$

Note that the derivative of an analytic function can be obtained simply by differentiating $G(s)$ with respect to s . In this example,

$$\frac{d}{ds} \left(\frac{1}{s+1} \right) = -\frac{1}{(s+1)^2}$$

Points in the s plane at which the function $G(s)$ is analytic are called *ordinary* points, while points in the s plane at which the function $G(s)$ is not analytic are called *singular* points. Singular points at which the function $G(s)$ or its derivatives approach infinity are called *poles*. In the previous example, $s = -1$ is a singular point and is a pole of the function $G(s)$.

If $G(s)$ approaches infinity as s approaches $-p$ and if the function

$$G(s)(s+p)^n, \quad \text{for } n = 1, 2, 3, \dots$$

has a finite, nonzero value at $s = -p$, then $s = -p$ is called a pole of order n . If $n = 1$, the pole is called a simple pole. If $n = 2, 3, \dots$, the pole is called a second-order pole, a third-order pole, and so on. Points at which the function $G(s)$ equals zero are called *zeros*.

To illustrate, consider the complex function

$$G(s) = \frac{K(s+2)(s+10)}{s(s+1)(s+5)(s+15)^2}$$

$G(s)$ has zeros at $s = -2, s = -10$, simple poles at $s = 0, s = -1, s = -5$, and a double pole (multiple pole of order 2) at $s = -15$. Note that $G(s)$ becomes zero at $s = \infty$. Since for large values of s

$$G(s) \doteq \frac{K}{s^3}$$

$G(s)$ possesses a triple zero (multiple zero of order 3) at $s = \infty$. If points at infinity are included, $G(s)$ has the same number of poles as zeros. To summarize, $G(s)$ has five zeros ($s = -2, s = -10, s = \infty, s = \infty, s = \infty$) and five poles ($s = 0, s = -1, s = -5, s = -15, s = -15$).

Euler's theorem. The power series expansions of $\cos \theta$ and $\sin \theta$ are, respectively,

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots$$

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots$$

And so

$$\cos \theta + j \sin \theta = 1 + (j\theta) + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \dots$$

Since

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

we see that

$$\cos \theta + j \sin \theta = e^{j\theta} \quad (2-1)$$

This is known as *Euler's theorem*.

By using Euler's theorem, we can express sine and cosine in terms of an exponential function. Noting that $e^{-j\theta}$ is the complex conjugate of $e^{j\theta}$ and that

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{-j\theta} = \cos \theta - j \sin \theta$$

we find, after adding and subtracting these two equations, that

$$\cos \theta = \frac{1}{2} (e^{j\theta} + e^{-j\theta}) \quad (2-2)$$

$$\sin \theta = \frac{1}{2j} (e^{j\theta} - e^{-j\theta}) \quad (2-3)$$

2-3 LAPLACE TRANSFORMATION

We shall first present a definition of the Laplace transformation and a brief discussion of the condition for the existence of the Laplace transform and then provide examples for illustrating the derivation of Laplace transforms of several common functions.

Let us define

$f(t)$ = a function of time t such that $f(t) = 0$ for $t < 0$

s = a complex variable

$\mathcal{L}$ = an operational symbol indicating that the quantity that it prefixes is to be transformed by the Laplace integral $\int_0^\infty e^{-st} dt$

$F(s)$ = Laplace transform of $f(t)$

Then the Laplace transform of $f(t)$ is given by

$$\mathcal{L}[f(t)] = F(s) = \int_0^\infty e^{-st} dt[f(t)] = \int_0^\infty f(t)e^{-st} dt$$

The reverse process of finding the time function $f(t)$ from the Laplace transform $F(s)$ is called the *inverse Laplace transformation*. The notation for the inverse Laplace transformation is $\mathcal{L}^{-1}$, and the inverse Laplace transform can be found from $F(s)$ by the following inversion integral:

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds, \quad \text{for } t > 0 \quad (2-4)$$

where c , the abscissa of convergence, is a real constant and is chosen larger than the real parts of all singular points of $F(s)$. Thus, the path of integration is parallel to the $j\omega$ axis and is displaced by the amount c from it. This path of integration is to the right of all singular points.

Evaluating the inversion integral appears complicated. In practice, we seldom use this integral for finding $f(t)$. There are simpler methods for finding $f(t)$. We shall discuss such simpler methods in Section 2-5.

It is noted that in this book the time function $f(t)$ is always assumed to be zero for negative time; that is,

$$f(t) = 0, \quad \text{for } t < 0$$

Existence of Laplace transform. The Laplace transform of a function $f(t)$ exists if the Laplace integral converges. The integral will converge if $f(t)$ is sectionally continuous in every finite interval in the range $t > 0$ and if it is of exponential order as t approaches infinity. A function $f(t)$ is said to be of exponential order if a real, positive constant σ exists such that the function

$$e^{-\sigma t}|f(t)|$$

approaches zero as t approaches infinity. If the limit of the function $e^{-\sigma t}|f(t)|$ approaches zero for σ greater than σ_c and the limit approaches infinity for σ less than σ_c , the value σ_c is called the *abscissa of convergence*.

For the function $f(t) = Ae^{-\alpha t}$

$$\lim_{t \rightarrow \infty} e^{-\sigma t}|Ae^{-\alpha t}|$$

approaches zero if $\sigma > -\alpha$. The abscissa of convergence in this case is $\sigma_c = -\alpha$. The integral $\int_0^\infty f(t)e^{-st} dt$ converges only if σ , the real part of s , is greater than the abscissa of convergence σ_c . Thus the operator s must be chosen as a constant such that this integral converges.

In terms of the poles of the function $F(s)$, the abscissa of convergence σ_c corresponds to the real part of the pole located farthest to the right in the s plane. For example, for the following function $F(s)$,

$$F(s) = \frac{K(s+3)}{(s+1)(s+2)}$$

the abscissa of convergence σ_c is equal to -1 . It can be seen that for such functions as t , $\sin \omega t$, and $t \sin \omega t$ the abscissa of convergence is equal to zero. For functions like e^{-ct} , te^{-ct} , $e^{-ct} \sin \omega t$, and so on, the abscissa of convergence is equal to $-c$. For functions that increase faster than the exponential function, however, it is impossible to find suitable values of the abscissa of convergence. Therefore, such functions as e^{t^2} and te^{t^2} do not possess Laplace transforms.

The reader should be cautioned that although e^{t^2} (for $0 \leq t \leq \infty$) does not possess a Laplace transform, the time function defined by

$$\begin{aligned} f(t) &= e^{t^2}, & \text{for } 0 \leq t \leq T < \infty \\ &= 0, & \text{for } t < 0, T < t \end{aligned}$$

does possess a Laplace transform since $f(t) = e^{t^2}$ for only a limited time interval $0 \leq t \leq T$ and not for $0 \leq t \leq \infty$. Such a signal can be physically generated. Note that the signals that we can physically generate always have corresponding Laplace transforms.

If a function $f(t)$ has a Laplace transform, then the Laplace transform of $Af(t)$, where A is a constant, is given by

$$\mathcal{L}[Af(t)] = A\mathcal{L}[f(t)]$$

This is obvious from the definition of the Laplace transform. Similarly, if functions $f_1(t)$ and $f_2(t)$ have Laplace transforms, then the Laplace transform of the function $f_1(t) + f_2(t)$ is given by

$$\mathcal{L}[f_1(t) + f_2(t)] = \mathcal{L}[f_1(t)] + \mathcal{L}[f_2(t)]$$

Again the proof of this relationship is evident from the definition of the Laplace transform.

In what follows, we derive Laplace transforms of a few commonly encountered functions.

Exponential function. Consider the exponential function

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= Ae^{-\alpha t}, & \text{for } t \geq 0 \end{aligned}$$

where A and α are constants. The Laplace transform of this exponential function can be obtained as follows:

$$\mathcal{L}[Ae^{-\alpha t}] = \int_0^{\infty} Ae^{-\alpha t} e^{-st} dt = A \int_0^{\infty} e^{-(\alpha+s)t} dt = \frac{A}{s + \alpha}$$

It is seen that the exponential function produces a pole in the complex plane.

In deriving the Laplace transform of $f(t) = Ae^{-\alpha t}$, we required that the real part of s be greater than $-\alpha$ (the abscissa of convergence). A question may immediately arise as to whether or not the Laplace transform thus obtained is valid in the range where $\sigma < -\alpha$ in the s plane. To answer this question, we must resort to the theory of complex variables. In the theory of complex variables, there is a theorem known as the analytic extension theorem. It states that, if two analytic functions are equal for a finite length along any arc in a region in which both are analytic, then they are equal everywhere in the region. The arc of equality is usually the real axis or a portion of it. By using this theorem the form of $F(s)$ determined by an integration in which s is allowed to have any real positive value greater than the abscissa of convergence holds for any complex values of s at which $F(s)$ is analytic. Thus, although we require the real part of s to be greater than the abscissa of convergence to make the $\int_0^{\infty} f(t)e^{-st} dt$ absolutely convergent, once the Laplace transform $F(s)$ is obtained, $F(s)$ can be considered valid throughout the entire s plane except at the poles of $F(s)$.

Step function. Consider the step function

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= A, & \text{for } t > 0 \end{aligned}$$

where A is a constant. Note that it is a special case of the exponential function $Ae^{-\alpha t}$, where $\alpha = 0$. The step function is undefined at $t = 0$. Its Laplace transform is given by

$$\mathcal{L}[A] = \int_0^{\infty} Ae^{-st} dt = \frac{A}{s}$$

In performing this integration, we assumed that the real part of s was greater than zero (the abscissa of convergence) and therefore that $\lim_{t \rightarrow \infty} e^{-st}$ was zero. As stated

earlier, the Laplace transform thus obtained is valid in the entire s plane except at the pole $s = 0$.

The step function whose height is unity is called *unit-step* function. The unit-step function that occurs at $t = t_0$ is frequently written as $1(t - t_0)$. The step function of height A that occurs at $t = 0$ can then be written as $f(t) = A1(t)$. The Laplace transform of the unit-step function, which is defined by

$$\begin{aligned} 1(t) &= 0, & \text{for } t < 0 \\ &= 1, & \text{for } t > 0 \end{aligned}$$

is $1/s$, or

$$\mathcal{L}[1(t)] = \frac{1}{s}$$

Physically, a step function occurring at $t = 0$ corresponds to a constant signal suddenly applied to the system at time t equals zero.

Ramp function. Consider the ramp function

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= At, & \text{for } t \geq 0 \end{aligned}$$

where A is a constant. The Laplace transform of this ramp function is obtained as

$$\begin{aligned} \mathcal{L}[At] &= \int_0^{\infty} Ate^{-st} dt = At \frac{e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} \frac{Ae^{-st}}{-s} dt \\ &= \frac{A}{s} \int_0^{\infty} e^{-st} dt = \frac{A}{s^2} \end{aligned}$$

Sinusoidal function. The Laplace transform of the sinusoidal function

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= A \sin \omega t, & \text{for } t \geq 0 \end{aligned}$$

where A and ω are constants, is obtained as follows. Referring to Equation (2-3), $\sin \omega t$ can be written

$$\sin \omega t = \frac{1}{2j} (e^{j\omega t} - e^{-j\omega t})$$

Hence

$$\begin{aligned} \mathcal{L}[A \sin \omega t] &= \frac{A}{2j} \int_0^{\infty} (e^{j\omega t} - e^{-j\omega t}) e^{-st} dt \\ &= \frac{A}{2j} \frac{1}{s - j\omega} - \frac{A}{2j} \frac{1}{s + j\omega} = \frac{A\omega}{s^2 + \omega^2} \end{aligned}$$

Similarly, the Laplace transform of $A \cos \omega t$ can be derived as follows:

$$\mathcal{L}[A \cos \omega t] = \frac{As}{s^2 + \omega^2}$$

Comments. The Laplace transform of any Laplace transformable function $f(t)$ can be found by multiplying $f(t)$ by e^{-st} and then integrating the product from $t = 0$ to $t = \infty$. Once we know the method of obtaining the Laplace transform, however, it is not necessary to derive the Laplace transform of $f(t)$ each time. Laplace transform tables can conveniently be used to find the transform of a given function $f(t)$. Table 2-1 shows Laplace transforms of time functions that will frequently appear in linear control systems analysis.

In the following discussion we present Laplace transforms of functions as well as theorems on the Laplace transformation that are useful in the study of linear control systems.

Translated function. Let us obtain the Laplace transform of the translated function $f(t - \alpha)1(t - \alpha)$, where $\alpha \geq 0$. This function is zero for $t < \alpha$. The functions $f(t)1(t)$ and $f(t - \alpha)1(t - \alpha)$ are shown in Figure 2-1.

By definition, the Laplace transform of $f(t - \alpha)1(t - \alpha)$ is

$$\mathcal{L}[f(t - \alpha)1(t - \alpha)] = \int_0^{\infty} f(t - \alpha)1(t - \alpha)e^{-st} dt$$

By changing the independent variable from t to τ , where $\tau = t - \alpha$, we obtain

$$\int_0^{\infty} f(t - \alpha)1(t - \alpha)e^{-st} dt = \int_{-\alpha}^{\infty} f(\tau)1(\tau)e^{-s(\tau+\alpha)} d\tau$$

Since in this book we always assume that $f(t) = 0$ for $t < 0$, $f(\tau)1(\tau) = 0$ for $\tau < 0$. Hence we can change the lower limit of integration from $-\alpha$ to 0. Thus

$$\begin{aligned} \int_{-\alpha}^{\infty} f(\tau)1(\tau)e^{-s(\tau+\alpha)} d\tau &= \int_0^{\infty} f(\tau)1(\tau)e^{-s(\tau+\alpha)} d\tau \\ &= \int_0^{\infty} f(\tau)e^{-s\tau}e^{-\alpha s} d\tau \\ &= e^{-\alpha s} \int_0^{\infty} f(\tau)e^{-s\tau} d\tau = e^{-\alpha s} F(s) \end{aligned}$$

where

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t)e^{-st} dt$$

And so

$$\mathcal{L}[f(t - \alpha)1(t - \alpha)] = e^{-\alpha s} F(s), \quad \text{for } \alpha \geq 0$$

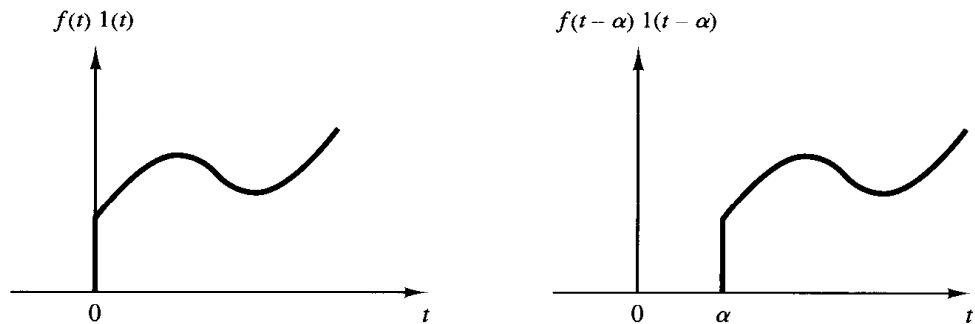


Figure 2-1
Function $f(t)1(t)$ and
translated function
 $f(t - \alpha)1(t - \alpha)$.

Table 2–1 Laplace Transform Pairs

	$f(t)$	$F(s)$
1	Unit impulse $\delta(t)$	1
2	Unit step $1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	$\frac{t^{n-1}}{(n-1)!} \quad (n = 1, 2, 3, \dots)$	$\frac{1}{s^n}$
5	$t^n \quad (n = 1, 2, 3, \dots)$	$\frac{n!}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	$\frac{1}{(n-1)!} t^{n-1} e^{-at} \quad (n = 1, 2, 3, \dots)$	$\frac{1}{(s+a)^n}$
9	$t^n e^{-at} \quad (n = 1, 2, 3, \dots)$	$\frac{n!}{(s+a)^{n+1}}$
10	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
11	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
12	$\sinh \omega t$	$\frac{\omega}{s^2 - \omega^2}$
13	$\cosh \omega t$	$\frac{s}{s^2 - \omega^2}$
14	$\frac{1}{a} (1 - e^{-at})$	$\frac{1}{s(s+a)}$
15	$\frac{1}{b-a} (e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
16	$\frac{1}{b-a} (be^{-bt} - ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
17	$\frac{1}{ab} \left[1 + \frac{1}{a-b} (be^{-at} - ae^{-bt}) \right]$	$\frac{1}{s(s+a)(s+b)}$

Table 2-1 (Continued)

18	$\frac{1}{a^2} (1 - e^{-at} - ate^{-at})$	$\frac{1}{s(s+a)^2}$
19	$\frac{1}{a^2} (at - 1 + e^{-at})$	$\frac{1}{s^2(s+a)}$
20	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
21	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$
22	$\frac{\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin \omega_n \sqrt{1-\xi^2} t$	$\frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$
23	$-\frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin(\omega_n \sqrt{1-\xi^2} t - \phi)$ $\phi = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$	$\frac{s}{s^2 + 2\xi\omega_n s + \omega_n^2}$
24	$1 - \frac{1}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin(\omega_n \sqrt{1-\xi^2} t + \phi)$ $\phi = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$	$\frac{\omega_n^2}{s(s^2 + 2\xi\omega_n s + \omega_n^2)}$
25	$1 - \cos \omega t$	$\frac{\omega^2}{s(s^2 + \omega^2)}$
26	$\omega t - \sin \omega t$	$\frac{\omega^3}{s^2(s^2 + \omega^2)}$
27	$\sin \omega t - \omega t \cos \omega t$	$\frac{2\omega^3}{(s^2 + \omega^2)^2}$
28	$\frac{1}{2\omega} t \sin \omega t$	$\frac{s}{(s^2 + \omega^2)^2}$
29	$t \cos \omega t$	$\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$
30	$\frac{1}{\omega_2^2 - \omega_1^2} (\cos \omega_1 t - \cos \omega_2 t) \quad (\omega_1^2 \neq \omega_2^2)$	$\frac{s}{(s^2 + \omega_1^2)(s^2 + \omega_2^2)}$
31	$\frac{1}{2\omega} (\sin \omega t + \omega t \cos \omega t)$	$\frac{s^2}{(s^2 + \omega^2)^2}$

This last equation states that the translation of the time function $f(t)1(t)$ by α (where $\alpha \geq 0$) corresponds to the multiplication of the transform $F(s)$ by $e^{-\alpha s}$.

Pulse function. Consider the pulse function

$$\begin{aligned} f(t) &= \frac{A}{t_0}, & \text{for } 0 < t < t_0 \\ &= 0, & \text{for } t < 0, t_0 < t \end{aligned}$$

where A and t_0 are constants.

The pulse function here may be considered a step function of height A/t_0 that begins at $t = 0$ and that is superimposed by a negative step function of height A/t_0 beginning at $t = t_0$; that is,

$$f(t) = \frac{A}{t_0} 1(t) - \frac{A}{t_0} 1(t - t_0)$$

Then the Laplace transform of $f(t)$ is obtained as

$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}\left[\frac{A}{t_0} 1(t)\right] - \mathcal{L}\left[\frac{A}{t_0} 1(t - t_0)\right] \\ &= \frac{A}{t_0 s} - \frac{A}{t_0 s} e^{-st_0} \\ &= \frac{A}{t_0 s} (1 - e^{-st_0}) \end{aligned} \tag{2-5}$$

Impulse function. The impulse function is a special limiting case of the pulse function. Consider the impulse function

$$\begin{aligned} g(t) &= \lim_{t_0 \rightarrow 0} \frac{A}{t_0}, & \text{for } 0 < t < t_0 \\ &= 0, & \text{for } t < 0, t_0 < t \end{aligned}$$

Since the height of the impulse function is A/t_0 and the duration is t_0 , the area under the impulse is equal to A . As the duration t_0 approaches zero, the height A/t_0 approaches infinity, but the area under the impulse remains equal to A . Note that the magnitude of the impulse is measured by its area.

Referring to Equation (2-5), the Laplace transform of this impulse function is shown to be

$$\begin{aligned} \mathcal{L}[g(t)] &= \lim_{t_0 \rightarrow 0} \left[\frac{A}{t_0 s} (1 - e^{-st_0}) \right] \\ &= \lim_{t_0 \rightarrow 0} \frac{\frac{d}{dt_0} [A(1 - e^{-st_0})]}{\frac{d}{dt_0} (t_0 s)} = \frac{As}{s} = A \end{aligned}$$

Thus the Laplace transform of the impulse function is equal to the area under the impulse.

The impulse function whose area is equal to unity is called the *unit-impulse function* or the *Dirac delta function*. The unit-impulse function occurring at $t = t_0$ is usually denoted by $\delta(t - t_0)$. $\delta(t - t_0)$ satisfies the following:

$$\begin{aligned}\delta(t - t_0) &= 0, & \text{for } t \neq t_0 \\ \delta(t - t_0) &= \infty, & \text{for } t = t_0 \\ \int_{-\infty}^{\infty} \delta(t - t_0) dt &= 1\end{aligned}$$

It should be mentioned that an impulse that has an infinite magnitude and zero duration is mathematical fiction and does not occur in physical systems. If, however, the magnitude of a pulse input to a system is very large and its duration is very short compared to the system time constants, then we can approximate the pulse input by an impulse function. For instance, if a force or torque input $f(t)$ is applied to a system for a very short time duration, $0 < t < t_0$, where the magnitude of $f(t)$ is sufficiently large so that the integral $\int_0^{t_0} f(t) dt$ is not negligible, then this input can be considered an impulse input. (Note that when we describe the impulse input the area or magnitude of the impulse is most important, but the exact shape of the impulse is usually immaterial.) The impulse input supplies energy to the system in an infinitesimal time.

The concept of the impulse function is quite useful in differentiating discontinuous functions. The unit-impulse function $\delta(t - t_0)$ can be considered the derivative of the unit-step function $1(t - t_0)$ at the point of discontinuity $t = t_0$ or

$$\delta(t - t_0) = \frac{d}{dt} 1(t - t_0)$$

Conversely, if the unit-impulse function $\delta(t - t_0)$ is integrated, the result is the unit-step function $1(t - t_0)$. With the concept of the impulse function we can differentiate a function containing discontinuities, giving impulses, the magnitudes of which are equal to the magnitude of each corresponding discontinuity.

Multiplication of $f(t)$ by $e^{-\alpha t}$. If $f(t)$ is Laplace transformable, its Laplace transform being $F(s)$, then the Laplace transform of $e^{-\alpha t}f(t)$ is obtained as

$$\mathcal{L}[e^{-\alpha t}f(t)] = \int_0^{\infty} e^{-\alpha t}f(t)e^{-st} dt = F(s + \alpha) \quad (2-6)$$

We see that the multiplication of $f(t)$ by $e^{-\alpha t}$ has the effect of replacing s by $(s + \alpha)$ in the Laplace transform. Conversely, changing s to $(s + \alpha)$ is equivalent to multiplying $f(t)$ by $e^{-\alpha t}$. (Note that α may be real or complex.)

The relationship given by Equation (2-6) is useful in finding the Laplace transforms of such functions as $e^{-\alpha t} \sin \omega t$ and $e^{-\alpha t} \cos \omega t$. For instance, since

$$\mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2} = F(s), \quad \mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2} = G(s)$$

it follows from Equation (2-6) that the Laplace transforms of $e^{-\alpha t} \sin \omega t$ and $e^{-\alpha t} \cos \omega t$ are given, respectively, by

$$\mathcal{L}[e^{-\alpha t} \sin \omega t] = F(s + \alpha) = \frac{\omega}{(s + \alpha)^2 + \omega^2}$$

$$\mathcal{L}[e^{-\alpha t} \cos \omega t] = G(s + \alpha) = \frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$$

Change of time scale. In analyzing physical systems, it is sometimes desirable to change the time scale or normalize a given time function. The result obtained in terms of normalized time is useful because it can be applied directly to different systems having similar mathematical equations.

If t is changed into t/α , where α is a positive constant, then the function $f(t)$ is changed into $f(t/\alpha)$. If we denote the Laplace transform of $f(t)$ by $F(s)$, then the Laplace transform of $f(t/\alpha)$ may be obtained as follows:

$$\mathcal{L}\left[f\left(\frac{t}{\alpha}\right)\right] = \int_0^{\infty} f\left(\frac{t}{\alpha}\right) e^{-st} dt$$

Letting $t/\alpha = t_1$ and $\alpha s = s_1$, we obtain

$$\begin{aligned} \mathcal{L}\left[f\left(\frac{t}{\alpha}\right)\right] &= \int_0^{\infty} f(t_1) e^{-s_1 t_1} d(\alpha t_1) \\ &= \alpha \int_0^{\infty} f(t_1) e^{-s_1 t_1} dt_1 \\ &= \alpha F(s_1) \end{aligned}$$

or

$$\mathcal{L}\left[f\left(\frac{t}{\alpha}\right)\right] = \alpha F(\alpha s)$$

As an example, consider $f(t) = e^{-t}$ and $f(t/5) = e^{-0.2t}$. We obtain

$$\mathcal{L}[f(t)] = \mathcal{L}[e^{-t}] = F(s) = \frac{1}{s + 1}$$

Hence

$$\mathcal{L}\left[f\left(\frac{t}{5}\right)\right] = \mathcal{L}[e^{-0.2t}] = 5F(5s) = \frac{5}{5s + 1}$$

This result can be verified easily by taking the Laplace transform of $e^{-0.2t}$ directly as follows:

$$\mathcal{L}[e^{-0.2t}] = \frac{1}{s + 0.2} = \frac{5}{5s + 1}$$

Comments on the lower limit of the Laplace integral. In some cases, $f(t)$ possesses an impulse function at $t = 0$. Then the lower limit of the Laplace integral must be clearly specified as to whether it is 0^- or 0^+ , since the Laplace transforms of $f(t)$ differ

for these two lower limits. If such a distinction of the lower limit of the Laplace integral is necessary, we use the notations

$$\mathcal{L}_+[f(t)] = \int_{0+}^{\infty} f(t)e^{-st} dt$$

$$\mathcal{L}_-[f(t)] = \int_{0-}^{\infty} f(t)e^{-st} dt = \mathcal{L}_+[f(t)] + \int_{0-}^{0+} f(t)e^{-st} dt$$

If $f(t)$ involves an impulse function at $t = 0$, then

$$\mathcal{L}_+[f(t)] \neq \mathcal{L}_-[f(t)]$$

since

$$\int_{0-}^{0+} f(t)e^{-st} dt \neq 0$$

for such a case. Obviously, if $f(t)$ does not possess an impulse function at $t = 0$ (that is, if the function to be transformed is finite between $t = 0-$ and $t = 0+$), then

$$\mathcal{L}_+[f(t)] = \mathcal{L}_-[f(t)]$$

2-4 LAPLACE TRANSFORM THEOREMS

This section presents several theorems on Laplace transformation that are important in control engineering.

Real differentiation theorem. The Laplace transform of the derivative of a function $f(t)$ is given by

$$\mathcal{L}\left[\frac{d}{dt} f(t)\right] = sF(s) - f(0) \quad (2-7)$$

where $f(0)$ is the initial value of $f(t)$ evaluated at $t = 0$.

For a given function $f(t)$, the values of $f(0+)$ and $f(0-)$ may be the same or different, as illustrated in Figure 2-2. The distinction between $f(0+)$ and $f(0-)$ is important when $f(t)$ has a discontinuity at $t = 0$ because in such a case $df(t)/dt$ will involve an impulse function at $t = 0$. If $f(0+) \neq f(0-)$, Equation (2-7) must be modified to

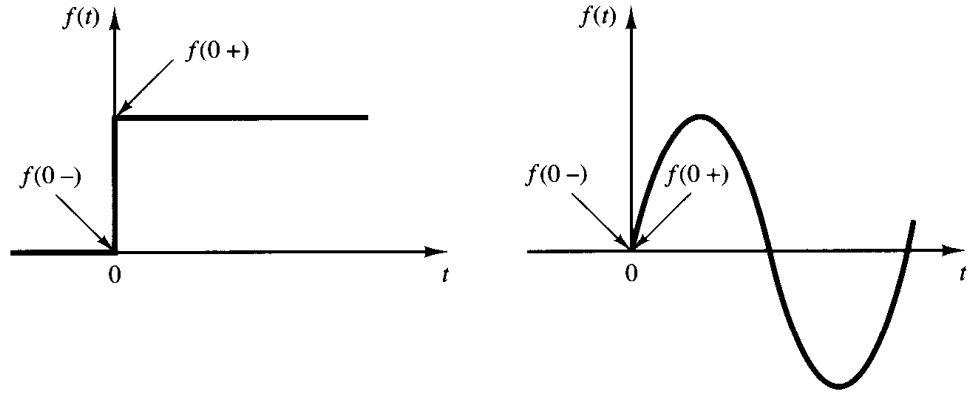
$$\begin{aligned} \mathcal{L}_+\left[\frac{d}{dt} f(t)\right] &= sF(s) - f(0+) \\ \mathcal{L}_-\left[\frac{d}{dt} f(t)\right] &= sF(s) - f(0-) \end{aligned}$$

To prove the real differentiation theorem, Equation (2-7), we proceed as follows. Integrating the Laplace integral by parts gives

$$\int_0^{\infty} f(t)e^{-st} dt = f(t) \frac{e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} \left[\frac{d}{dt} f(t)\right] \frac{e^{-st}}{-s} dt$$

Figure 2-2

Step function and sine function indicating initial values at $t = 0^-$ and $t = 0^+$.



Hence

$$F(s) = \frac{f(0)}{s} + \frac{1}{s} \mathcal{L} \left[\frac{d}{dt} f(t) \right]$$

It follows that

$$\mathcal{L} \left[\frac{d}{dt} f(t) \right] = sF(s) - f(0)$$

Similarly, we obtain the following relationship for the second derivative of $f(t)$:

$$\mathcal{L} \left[\frac{d^2}{dt^2} f(t) \right] = s^2 F(s) - sf(0) - \dot{f}(0)$$

where $\dot{f}(0)$ is the value of $df(t)/dt$ evaluated at $t = 0$. To derive this equation, define

$$\frac{d}{dt} f(t) = g(t)$$

Then

$$\begin{aligned} \mathcal{L} \left[\frac{d^2}{dt^2} f(t) \right] &= \mathcal{L} \left[\frac{d}{dt} g(t) \right] = s\mathcal{L}[g(t)] - g(0) \\ &= s\mathcal{L} \left[\frac{d}{dt} f(t) \right] - \dot{f}(0) \\ &= s^2 F(s) - sf(0) - \dot{f}(0) \end{aligned}$$

Similarly, for the n th derivative of $f(t)$, we obtain

$$\mathcal{L} \left[\frac{d^n}{dt^n} f(t) \right] = s^n F(s) - s^{n-1} f(0) - s^{n-2} \dot{f}(0) - \dots - s^{(n-2)} \frac{d^{n-2} f(t)}{dt^{n-2}} \Big|_{t=0} - \frac{d^{n-1} f(t)}{dt^{n-1}} \Big|_{t=0}$$

where $f(0), \dot{f}(0), \dots, \frac{d^{n-1} f(t)}{dt^{n-1}} \Big|_{t=0}$ represent the values of $f(t), df(t)/dt, \dots, d^{n-1} f(t)/dt^{n-1}$, respectively, evaluated at $t = 0$. If the distinction between $\mathcal{L}_+$ and $\mathcal{L}_-$ is necessary, we substitute $t = 0^+$ or $t = 0^-$ into $f(t), df(t)/dt, \dots, d^{n-1} f(t)/dt^{n-1}$, depending on whether we take $\mathcal{L}_+$ or $\mathcal{L}_-$.

Note that, in order for Laplace transforms of derivatives of $f(t)$ to exist, $d^n f(t)/dt^n$ ($n = 1, 2, 3, \dots$) must be Laplace transformable.

Note also that, if all the initial values of $f(t)$ and its derivatives are equal to zero, then the Laplace transform of the n th derivative of $f(t)$ is given by $s^n F(s)$.

EXAMPLE 2-1

Consider the cosine function.

$$\begin{aligned} g(t) &= 0, & \text{for } t < 0 \\ &= \cos \omega t, & \text{for } t \geq 0 \end{aligned}$$

The Laplace transform of this cosine function can be obtained directly as in the case of the sinusoidal function considered earlier. The use of the real differentiation theorem, however, will be demonstrated here by deriving the Laplace transform of the cosine function from the Laplace transform of the sine function. If we define

$$\begin{aligned} f(t) &= 0, & \text{for } t < 0 \\ &= \sin \omega t, & \text{for } t \geq 0 \end{aligned}$$

then

$$\mathcal{L}[\sin \omega t] = F(s) = \frac{\omega}{s^2 + \omega^2}$$

The Laplace transform of the cosine function is obtained as

$$\begin{aligned} \mathcal{L}[\cos \omega t] &= \mathcal{L}\left[\frac{1}{\omega} \left(\frac{d}{dt} \sin \omega t\right)\right] = \frac{1}{\omega} [sF(s) - f(0)] \\ &= \frac{1}{\omega} \left[\frac{s\omega}{s^2 + \omega^2} - 0\right] = \frac{s}{s^2 + \omega^2} \end{aligned}$$

Final-value theorem. The final-value theorem relates the steady-state behavior of $f(t)$ to the behavior of $sF(s)$ in the neighborhood of $s = 0$. This theorem, however, applies if and only if $\lim_{t \rightarrow \infty} f(t)$ exists [which means that $f(t)$ settles down to a definite value for $t \rightarrow \infty$]. If all poles of $sF(s)$ lie in the left half s plane, $\lim_{t \rightarrow \infty} f(t)$ exists. But if $sF(s)$ has poles on the imaginary axis or in the right half s plane, $f(t)$ will contain oscillating or exponentially increasing time functions, respectively, and $\lim_{t \rightarrow \infty} f(t)$ will not exist. The final-value theorem does not apply in such cases. For instance, if $f(t)$ is the sinusoidal function $\sin \omega t$, $sF(s)$ has poles at $s = \pm j\omega$ and $\lim_{t \rightarrow \infty} f(t)$ does not exist. Therefore, this theorem is not applicable to such a function.

The final-value theorem may be stated as follows. If $f(t)$ and $df(t)/dt$ are Laplace transformable, if $F(s)$ is the Laplace transform of $f(t)$, and if $\lim_{t \rightarrow \infty} f(t)$ exists, then

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

To prove the theorem, we let s approach zero in the equation for the Laplace transform of the derivative of $f(t)$ or

$$\lim_{s \rightarrow 0} \int_0^{\infty} \left[\frac{d}{dt} f(t) \right] e^{-st} dt = \lim_{s \rightarrow 0} [sF(s) - f(0)]$$

Since $\lim_{s \rightarrow 0} e^{-st} = 1$, we obtain

$$\begin{aligned} \int_0^{\infty} \left[\frac{d}{dt} f(t) \right] dt &= f(t) \Big|_0^{\infty} = f(\infty) - f(0) \\ &= \lim_{s \rightarrow 0} sF(s) - f(0) \end{aligned}$$

from which

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

The final-value theorem states that the steady-state behavior of $f(t)$ is the same as the behavior of $sF(s)$ in the neighborhood of $s = 0$. Thus, it is possible to obtain the value of $f(t)$ at $t = \infty$ directly from $F(s)$.

EXAMPLE 2-2

Given

$$\mathcal{L}[f(t)] = F(s) = \frac{1}{s(s+1)}$$

what is $\lim_{t \rightarrow \infty} f(t)$?

Since the pole of $sF(s) = 1/(s+1)$ lies in the left half s plane, $\lim_{t \rightarrow \infty} f(t)$ exists. So the final-value theorem is applicable in this case.

$$\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} \frac{s}{s(s+1)} = \lim_{s \rightarrow 0} \frac{1}{s+1} = 1$$

In fact, this result can easily be verified, since

$$f(t) = 1 - e^{-t}, \quad \text{for } t \geq 0$$

Initial-value theorem. The initial-value theorem is the counterpart of the final-value theorem. By using this theorem, we are able to find the value of $f(t)$ at $t = 0+$ directly from the Laplace transform of $f(t)$. The initial-value theorem does not give the value of $f(t)$ at exactly $t = 0$ but at a time slightly greater than zero.

The initial-value theorem may be stated as follows: If $f(t)$ and $df(t)/dt$ are both Laplace transformable and if $\lim_{s \rightarrow \infty} sF(s)$ exists, then

$$f(0+) = \lim_{s \rightarrow \infty} sF(s)$$

To prove this theorem, we use the equation for the $\mathcal{L}_+$ transform of $df(t)/dt$:

$$\mathcal{L}_+ \left[\frac{d}{dt} f(t) \right] = sF(s) - f(0+)$$

For the time interval $0+ \leq t \leq \infty$, as s approaches infinity, e^{-st} approaches zero. (Note that we must use $\mathcal{L}_+$ rather than $\mathcal{L}_-$ for this condition.) And so

$$\lim_{s \rightarrow \infty} \int_{0+}^{\infty} \left[\frac{d}{dt} f(t) \right] e^{-st} dt = \lim_{s \rightarrow \infty} [sF(s) - f(0+)] = 0$$

or

$$f(0+) = \lim_{s \rightarrow \infty} sF(s)$$

In applying the initial-value theorem, we are not limited as to the locations of the poles of $sF(s)$. Thus the initial-value theorem is valid for the sinusoidal function.

It should be noted that the initial-value theorem and the final-value theorem provide a convenient check on the solution, since they enable us to predict the system behavior in the time domain without actually transforming functions in s back to time functions.

Real-integration theorem. If $f(t)$ is of exponential order, then the Laplace transform of $\int f(t) dt$ exists and is given by

$$\mathcal{L}\left[\int f(t) dt\right] = \frac{F(s)}{s} + \frac{f^{-1}(0)}{s} \quad (2-8)$$

where $F(s) = \mathcal{L}[f(t)]$ and $f^{-1}(0) = \int f(t) dt$, evaluated at $t = 0$.

Note that if $f(t)$ involves an impulse function at $t = 0$, then $f^{-1}(0+) \neq f^{-1}(0-)$. So if $f(t)$ involves an impulse function at $t = 0$, we must modify Equation (2-8) as follows:

$$\begin{aligned} \mathcal{L}_+\left[\int f(t) dt\right] &= \frac{F(s)}{s} + \frac{f^{-1}(0+)}{s} \\ \mathcal{L}_-\left[\int f(t) dt\right] &= \frac{F(s)}{s} + \frac{f^{-1}(0-)}{s} \end{aligned}$$

The real-integration theorem given by Equation (2-8) can be proved in the following way. Integration by parts yields

$$\begin{aligned} \mathcal{L}\left[\int f(t) dt\right] &= \int_0^\infty \left[\int f(t) dt\right] e^{-st} dt \\ &= \left[\int f(t) dt\right] \frac{e^{-st}}{-s} \Big|_0^\infty - \int_0^\infty f(t) \frac{e^{-st}}{-s} dt \\ &= \frac{1}{s} \int f(t) dt \Big|_{t=0} + \frac{1}{s} \int_0^\infty f(t) e^{-st} dt \\ &= \frac{f^{-1}(0)}{s} + \frac{F(s)}{s} \end{aligned}$$

and the theorem is proved.

We see that integration in the time domain is converted into division in the s domain. If the initial value of the integral is zero, the Laplace transform of the integral of $f(t)$ is given by $F(s)/s$.

The preceding real-integration theorem given by Equation (2-8) can be modified slightly to deal with the definite integral of $f(t)$. If $f(t)$ is of exponential order, the Laplace transform of the definite integral $\int_0^t f(t) dt$ is given by

$$\mathcal{L}\left[\int_0^t f(t) dt\right] = \frac{F(s)}{s} \quad (2-9)$$

where $F(s) = \mathcal{L}[f(t)]$. This is also referred to as the real-integration theorem. Note that if $f(t)$ involves an impulse function at $t = 0$ then $\int_{0+}^t f(t) dt \neq \int_{0-}^t f(t) dt$, and the following distinction must be observed:

$$\begin{aligned} \mathcal{L}_+\left[\int_{0+}^t f(t) dt\right] &= \frac{\mathcal{L}_+[f(t)]}{s} \\ \mathcal{L}_-\left[\int_{0-}^t f(t) dt\right] &= \frac{\mathcal{L}_-[f(t)]}{s} \end{aligned}$$

To prove Equation (2-9), first note that

$$\int_0^t f(t) dt = \int f(t) dt - f^{-1}(0)$$

where $f^{-1}(0)$ is equal to $\int f(t) dt$ evaluated at $t = 0$ and is a constant. Hence

$$\mathcal{L}\left[\int_0^t f(t) dt\right] = \mathcal{L}\left[\int f(t) dt\right] - \mathcal{L}[f^{-1}(0)]$$

Noting that $f^{-1}(0)$ is a constant so that

$$\mathcal{L}[f^{-1}(0)] = \frac{f^{-1}(0)}{s}$$

we obtain

$$\mathcal{L}\left[\int_0^t f(t) dt\right] = \frac{F(s)}{s} + \frac{f^{-1}(0)}{s} - \frac{f^{-1}(0)}{s} = \frac{F(s)}{s}$$

Complex-differentiation theorem. If $f(t)$ is Laplace transformable, then, except at poles of $F(s)$,

$$\mathcal{L}[tf(t)] = -\frac{d}{ds} F(s)$$

where $F(s) = \mathcal{L}[f(t)]$. This is known as the complex-differentiation theorem. Also,

$$\mathcal{L}[t^2f(t)] = \frac{d^2}{ds^2} F(s)$$

In general,

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s), \quad \text{for } n = 1, 2, 3, \dots$$

To prove the complex-differentiation theorem, we proceed as follows:

$$\begin{aligned} \mathcal{L}[tf(t)] &= \int_0^\infty tf(t)e^{-st} dt = -\int_0^\infty f(t) \frac{d}{ds} (e^{-st}) dt \\ &= -\frac{d}{ds} \int_0^\infty f(t)e^{-st} dt = -\frac{d}{ds} F(s) \end{aligned}$$

Hence the theorem. Similarly, by defining $tf(t) = g(t)$, the result is

$$\begin{aligned} \mathcal{L}[t^2f(t)] &= \mathcal{L}[tg(t)] = -\frac{d}{ds} G(s) = -\frac{d}{ds} \left[-\frac{d}{ds} F(s) \right] \\ &= (-1)^2 \frac{d^2}{ds^2} F(s) = \frac{d^2}{ds^2} F(s) \end{aligned}$$

Repeating the same process, we obtain

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s), \quad \text{for } n = 1, 2, 3, \dots$$

Convolution integral. Consider the Laplace transform of

$$\int_0^t f_1(t - \tau) f_2(\tau) d\tau$$

This integral is often written as

$$f_1(t) * f_2(t)$$

The mathematical operation $f_1(t) * f_2(t)$ is called *convolution*. Note that if we put $t - \tau = \xi$, then

$$\begin{aligned} \int_0^t f_1(t - \tau) f_2(\tau) d\tau &= - \int_t^0 f_1(\xi) f_2(t - \xi) d\xi \\ &= \int_0^t f_1(\tau) f_2(t - \tau) d\tau \end{aligned}$$

Hence

$$\begin{aligned} f_1(t) * f_2(t) &= \int_0^t f_1(t - \tau) f_2(\tau) d\tau \\ &= \int_0^t f_1(\tau) f_2(t - \tau) d\tau \\ &= f_2(t) * f_1(t) \end{aligned}$$

If $f_1(t)$ and $f_2(t)$ are piecewise continuous and of exponential order, then the Laplace transform of

$$\int_0^t f_1(t - \tau) f_2(\tau) d\tau$$

can be obtained as follows:

$$\mathcal{L}\left[\int_0^t f_1(t - \tau) f_2(\tau) d\tau\right] = F_1(s) F_2(s) \quad (2-10)$$

where

$$F_1(s) = \int_0^\infty f_1(t) e^{-st} dt = \mathcal{L}[f_1(t)]$$

$$F_2(s) = \int_0^\infty f_2(t) e^{-st} dt = \mathcal{L}[f_2(t)]$$

To prove Equation (2-10) note that $f_1(t - \tau)1(t - \tau) = 0$ for $\tau > t$. Hence

$$\int_0^t f_1(t - \tau) f_2(\tau) d\tau = \int_0^\infty f_1(t - \tau) 1(t - \tau) f_2(\tau) d\tau$$

Then

$$\begin{aligned}\mathcal{L}\left[\int_0^t f_1(t-\tau)f_2(\tau) d\tau\right] &= \mathcal{L}\left[\int_0^\infty f_1(t-\tau)1(t-\tau)f_2(\tau) d\tau\right] \\ &= \int_0^\infty e^{-st}\left[\int_0^\infty f_1(t-\tau)1(t-\tau)f_2(\tau) d\tau\right] dt\end{aligned}$$

Substituting $t - \tau = \lambda$ in this last equation and changing the order of integration, which is valid in this case because $f_1(t)$ and $f_2(t)$ are Laplace transformable, we obtain

$$\begin{aligned}\mathcal{L}\left[\int_0^t f_1(t-\tau)f_2(\tau) d\tau\right] &= \int_0^\infty f_1(t-\tau)1(t-\tau)e^{-st} dt \int_0^\infty f_2(\tau) d\tau \\ &= \int_0^\infty f_1(\lambda)e^{-s(\lambda+\tau)} d\lambda \int_0^\infty f_2(\tau) d\tau \\ &= \int_0^\infty f_1(\lambda)e^{-s\lambda} d\lambda \int_0^\infty f_2(\tau)e^{-s\tau} d\tau \\ &= F_1(s)F_2(s)\end{aligned}$$

This last equation gives the Laplace transform of the convolution integral. Conversely, if the Laplace transform of a function is given by a product of two Laplace transform functions, $F_1(s)F_2(s)$, then the corresponding time function (the inverse Laplace transform) is given by the convolution integral $f_1(t)*f_2(t)$.

Laplace transform of product of two time functions. The Laplace transform of the product of two Laplace transformable functions $f(t)$ and $g(t)$ can be given by

$$\mathcal{L}[f(t)g(t)] = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(p)G(s-p) dp \quad (2-11)$$

To show this, we may proceed as follows: The Laplace transform of the product of $f(t)$ and $g(t)$ can be written as

$$\mathcal{L}[f(t)g(t)] = \int_0^\infty f(t)g(t)e^{-st} dt \quad (2-12)$$

Note that the inversion integral is

$$f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds, \quad \text{for } t > 0$$

where c is the abscissa of convergence for $F(s)$. Thus,

$$\mathcal{L}[f(t)g(t)] = \frac{1}{2\pi j} \int_0^\infty \int_{c-j\infty}^{c+j\infty} F(p)e^{pt} dp g(t)e^{-st} dt$$

Because of the uniform convergence of the integrals considered, we may invert the order of integration:

$$\mathcal{L}[f(t)g(t)] = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(p) dp \int_0^\infty g(t)e^{-(s-p)t} dt$$

Noting that

$$\int_0^\infty g(t)e^{-(s-p)t} dt = G(s-p)$$

we obtain

$$\mathcal{L}[f(t)g(t)] = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(p)G(s-p) dp \quad (2-13)$$

Summary. Table 2-2 summarizes properties and theorems of the Laplace transforms. Most of them have been derived or proved in this section.

2-5 INVERSE LAPLACE TRANSFORMATION

As noted earlier, the inverse Laplace transform can be obtained by use of the inversion integral given by Equation (2-4). However, the inversion integral is complicated and, therefore, its use is not recommended for finding inverse Laplace transforms of commonly encountered functions in control engineering.

A convenient method for obtaining inverse Laplace transforms is to use a table of Laplace transforms. In this case, the Laplace transform must be in a form immediately recognizable in such a table. Quite often the function in question may not appear in tables of Laplace transforms available to the engineer. If a particular transform $F(s)$ cannot be found in a table, then we may expand it into partial fractions and write $F(s)$ in terms of simple functions of s for which the inverse Laplace transforms are already known.

Note that these simpler methods for finding inverse Laplace transforms are based on the fact that the unique correspondence of a time function and its inverse Laplace transform holds for any continuous time function.

Partial-fraction expansion method for finding inverse Laplace transforms. For problems in control systems analysis, $F(s)$, the Laplace transform of $f(t)$, frequently occurs in the form

$$F(s) = \frac{B(s)}{A(s)}$$

where $A(s)$ and $B(s)$ are polynomials in s . In the expansion of $F(s) = B(s)/A(s)$ into a partial-fraction form, it is important that the highest power of s in $A(s)$ be greater than the highest power of s in $B(s)$. If such is not the case, the numerator $B(s)$ must be divided by the denominator $A(s)$ in order to produce a polynomial in s plus a remainder (a ratio of polynomials in s whose numerator is of lower degree than the denominator).

Table 2–2 Properties of Laplace Transforms

1	$\mathcal{L}[Af(t)] = AF(s)$
2	$\mathcal{L}[f_1(t) \pm f_2(t)] = F_1(s) \pm F_2(s)$
3	$\mathcal{L}_\pm \left[\frac{d}{dt} f(t) \right] = sF(s) - f(0\pm)$
4	$\mathcal{L}_\pm \left[\frac{d^2}{dt^2} f(t) \right] = s^2 F(s) - sf(0\pm) - \dot{f}(0\pm)$
5	$\mathcal{L}_\pm \left[\frac{d^n}{dt^n} f(t) \right] = s^n F(s) - \sum_{k=1}^n s^{n-k} f^{(k-1)}(0\pm)$ where $f^{(k-1)}(t) = \frac{d^{k-1}}{dt^{k-1}} f(t)$
6	$\mathcal{L}_\pm \left[\int f(t) dt \right] = \frac{F(s)}{s} + \frac{1}{s} \left[\int f(t) dt \right]_{t=0\pm}$
7	$\mathcal{L}_\pm \left[\int \cdots \int f(t)(dt)^n \right] = \frac{F(s)}{s^n} + \sum_{k=1}^n \frac{1}{s^{n-k+1}} \left[\int \cdots \int f(t)(dt)^k \right]_{t=0\pm}$
8	$\mathcal{L} \left[\int_0^t f(t) dt \right] = \frac{F(s)}{s}$
9	$\int_0^\infty f(t) dt = \lim_{s \rightarrow 0} F(s) \quad \text{if } \int_0^\infty f(t) dt \text{ exists}$
10	$\mathcal{L}[e^{-at} f(t)] = F(s + a)$
11	$\mathcal{L}[f(t - \alpha)1(t - \alpha)] = e^{-as} F(s) \quad \alpha \geq 0$
12	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
13	$\mathcal{L}[t^2 f(t)] = \frac{d^2}{ds^2} F(s)$
14	$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s) \quad n = 1, 2, 3, \dots$
15	$\mathcal{L} \left[\frac{1}{t} f(t) \right] = \int_s^\infty F(s) ds \quad \text{if } \lim_{t \rightarrow 0} \frac{1}{t} f(t) \text{ exists}$
16	$\mathcal{L} \left[f \left(\frac{t}{a} \right) \right] = aF(as)$
17	$\mathcal{L} \left[\int_0^t f_1(t - \tau) f_2(\tau) d\tau \right] = F_1(s) F_2(s)$
18	$\mathcal{L}[f(t)g(t)] = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(p)G(s - p) dp$

If $F(s)$ is broken up into components,

$$F(s) = F_1(s) + F_2(s) + \cdots + F_n(s)$$

and if the inverse Laplace transforms of $F_1(s), F_2(s), \dots, F_n(s)$ are readily available, then

$$\begin{aligned}\mathcal{L}^{-1}[F(s)] &= \mathcal{L}^{-1}[F_1(s)] + \mathcal{L}^{-1}[F_2(s)] + \cdots + \mathcal{L}^{-1}[F_n(s)] \\ &= f_1(t) + f_2(t) + \cdots + f_n(t)\end{aligned}$$

where $f_1(t), f_2(t), \dots, f_n(t)$ are the inverse Laplace transforms of $F_1(s), F_2(s), \dots, F_n(s)$, respectively. The inverse Laplace transform of $F(s)$ thus obtained is unique except possibly at points where the time function is discontinuous. Whenever the time function is continuous, the time function $f(t)$ and its Laplace transform $F(s)$ have a one-to-one correspondence.

The advantage of the partial-fraction expansion approach is that the individual terms of $F(s)$, resulting from the expansion into partial-fraction form, are very simple functions of s ; consequently, it is not necessary to refer to a Laplace transform table if we memorize several simple Laplace transform pairs. It should be noted, however, that in applying the partial-fraction expansion technique in the search for the inverse Laplace transform of $F(s) = B(s)/A(s)$ the roots of the denominator polynomial $A(s)$ must be obtained in advance. That is, this method does not apply until the denominator polynomial has been factored.

Partial-fraction expansion when $F(s)$ involves distinct poles only. Consider $F(s)$ written in the factored form

$$F(s) = \frac{B(s)}{A(s)} = \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)}, \quad \text{for } m < n$$

where $p_1, p_2, \dots, p_n$ and $z_1, z_2, \dots, z_m$ are either real or complex quantities, but for each complex p_i or z_i there will occur the complex conjugate of p_i or z_i , respectively. If $F(s)$ involves distinct poles only, then it can be expanded into a sum of simple partial fractions as follows:

$$F(s) = \frac{B(s)}{A(s)} = \frac{a_1}{s + p_1} + \frac{a_2}{s + p_2} + \cdots + \frac{a_n}{s + p_n} \quad (2-14)$$

where a_k ($k = 1, 2, \dots, n$) are constants. The coefficient a_k is called the *residue* at the pole at $s = -p_k$. The value of a_k can be found by multiplying both sides of Equation (2-14) by $(s + p_k)$ and letting $s = -p_k$, which gives

$$\begin{aligned}\left[(s + p_k) \frac{B(s)}{A(s)}\right]_{s=-p_k} &= \left[\frac{a_1}{s + p_1}(s + p_k) + \frac{a_2}{s + p_2}(s + p_k) \right. \\ &\quad \left. + \cdots + \frac{a_k}{s + p_k}(s + p_k) + \cdots + \frac{a_n}{s + p_n}(s + p_k)\right]_{s=-p_k} \\ &= a_k\end{aligned}$$

We see that all the expanded terms drop out with the exception of a_k . Thus the residue a_k is found from

$$a_k = \left[(s + p_k) \frac{B(s)}{A(s)} \right]_{s=-p_k} \quad (2-15)$$

Note that, since $f(t)$ is a real function of time, if p_1 and p_2 are complex conjugates, then the residues a_1 and a_2 are also complex conjugates. Only one of the conjugates, a_1 or a_2 , needs to be evaluated because the other is known automatically.

Since

$$\mathcal{L}^{-1} \left[\frac{a_k}{s + p_k} \right] = a_k e^{-p_k t}$$

$f(t)$ is obtained as

$$f(t) = \mathcal{L}^{-1}[F(s)] = a_1 e^{-p_1 t} + a_2 e^{-p_2 t} + \cdots + a_n e^{-p_n t}, \quad \text{for } t \geq 0$$

EXAMPLE 2-3 Find the inverse Laplace transform of

$$F(s) = \frac{s + 3}{(s + 1)(s + 2)}$$

The partial-fraction expansion of $F(s)$ is

$$F(s) = \frac{s + 3}{(s + 1)(s + 2)} = \frac{a_1}{s + 1} + \frac{a_2}{s + 2}$$

where a_1 and a_2 are found by using Equation (2-15):

$$\begin{aligned} a_1 &= \left[(s + 1) \frac{s + 3}{(s + 1)(s + 2)} \right]_{s=-1} = \left[\frac{s + 3}{s + 2} \right]_{s=-1} = 2 \\ a_2 &= \left[(s + 2) \frac{s + 3}{(s + 1)(s + 2)} \right]_{s=-2} = \left[\frac{s + 3}{s + 1} \right]_{s=-2} = -1 \end{aligned}$$

Thus

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}[F(s)] \\ &= \mathcal{L}^{-1} \left[\frac{2}{s + 1} \right] + \mathcal{L}^{-1} \left[\frac{-1}{s + 2} \right] \\ &= 2e^{-t} - e^{-2t}, \quad \text{for } t \geq 0 \end{aligned}$$

EXAMPLE 2-4 Obtain the inverse Laplace transform of

$$G(s) = \frac{s^3 + 5s^2 + 9s + 7}{(s + 1)(s + 2)}$$

Here, since the degree of the numerator polynomial is higher than that of the denominator polynomial, we must divide the numerator by the denominator.

$$G(s) = s + 2 + \frac{s + 3}{(s + 1)(s + 2)}$$

Note that the Laplace transform of the unit-impulse function $\delta(t)$ is 1 and that the Laplace transform of $d\delta(t)/dt$ is s . The third term on the right-hand side of this last equation is $F(s)$ in Example 2-3. So the inverse Laplace transform of $G(s)$ is given as

$$g(t) = \frac{d}{dt} \delta(t) + 2\delta(t) + 2e^{-t} - e^{-2t}, \quad \text{for } t \geq 0-$$

EXAMPLE 2-5 Find the inverse Laplace transform of

$$F(s) = \frac{2s + 12}{s^2 + 2s + 5}$$

Notice that the denominator polynomial can be factored as

$$s^2 + 2s + 5 = (s + 1 + j2)(s + 1 - j2)$$

If the function $F(s)$ involves a pair of complex-conjugate poles, it is convenient not to expand $F(s)$ into the usual partial fractions but to expand it into the sum of a damped sine and a damped cosine function.

Noting that $s^2 + 2s + 5 = (s + 1)^2 + 2^2$ and referring to the Laplace transforms of $e^{-\alpha t} \sin \omega t$ and $e^{-\alpha t} \cos \omega t$, rewritten thus,

$$\mathcal{L}[e^{-\alpha t} \sin \omega t] = \frac{\omega}{(s + \alpha)^2 + \omega^2}$$

$$\mathcal{L}[e^{-\alpha t} \cos \omega t] = \frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$$

the given $F(s)$ can be written as a sum of a damped sine and a damped cosine function.

$$\begin{aligned} F(s) &= \frac{2s + 12}{s^2 + 2s + 5} = \frac{10 + 2(s + 1)}{(s + 1)^2 + 2^2} \\ &= 5 \frac{2}{(s + 1)^2 + 2^2} + 2 \frac{s + 1}{(s + 1)^2 + 2^2} \end{aligned}$$

It follows that

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}[F(s)] \\ &= 5\mathcal{L}^{-1}\left[\frac{2}{(s + 1)^2 + 2^2}\right] + 2\mathcal{L}^{-1}\left[\frac{s + 1}{(s + 1)^2 + 2^2}\right] \\ &= 5e^{-t} \sin 2t + 2e^{-t} \cos 2t, \quad \text{for } t \geq 0 \end{aligned}$$

Partial-fraction expansion when $F(s)$ involves multiple poles. Instead of discussing the general case, we shall use an example to show how to obtain the partial-fraction expansion of $F(s)$. (See also Problem A-2-16.)

Consider the following $F(s)$:

$$F(s) = \frac{s^2 + 2s + 3}{(s + 1)^3}$$

The partial-fraction expansion of this $F(s)$ involves three terms,

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_1}{s + 1} + \frac{b_2}{(s + 1)^2} + \frac{b_3}{(s + 1)^3}$$